SIGCSE Virtual 2026
2nd ACM Virtual Global Computing Education Conference

Agents of Instruction

Co-Creating Learning Materials with AI Agents



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Welcome!

All materials for the workshop available at:
https://csedagentsofinstruction.github.io/

1. [15 min] Logistics; What are agents and how did we get here?
2. [30 min] How we (facilitators) use agents
3. [5 min] Responsible use
4. [30 min] Activity #1 + *break*
5. [15 min] Debrief
6. [30 min] Activity #2 + *break*
7. [15 min] Debrief
8. [10 min] Wrap-up and Takeaways



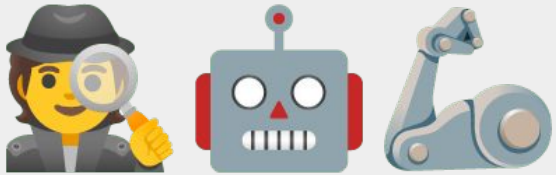
“Technology changes, standards are replaced, and pedagogical theory improves. In our experience, the only constant in development, adoption, and use of educational technology is eventual misery that equates with a need for change. Sisyphus, in Greek mythology, was cursed to forever push the same boulder up a hill – the rock was about to crest, it rolled back instead, and the toil started anew. This is also the fate of the computing educator striving to eternally align technology and pedagogy. However, the inevitable misery is not a reason to give up; we should keep on building systems and innovating pedagogy and technology alike.”

Today: Let's learn to use AI agents to do some of the cursed pushing.

Haaranen et al. "Decades of Striving for Pedagogical and Technological Alignment." Proceedings of the 23rd Koli Calling International Conference on Computing Education Research. 2023.



What are agents and how did we get here?



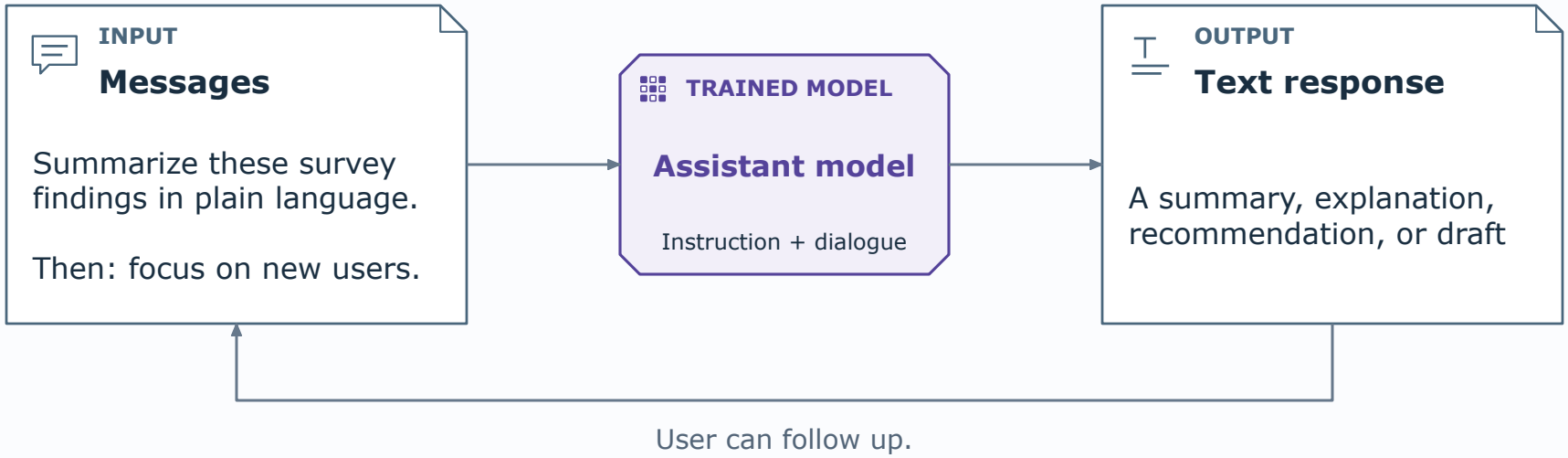
Text continuation

Early LLMs, 2017ish →



Instruction following and dialogue

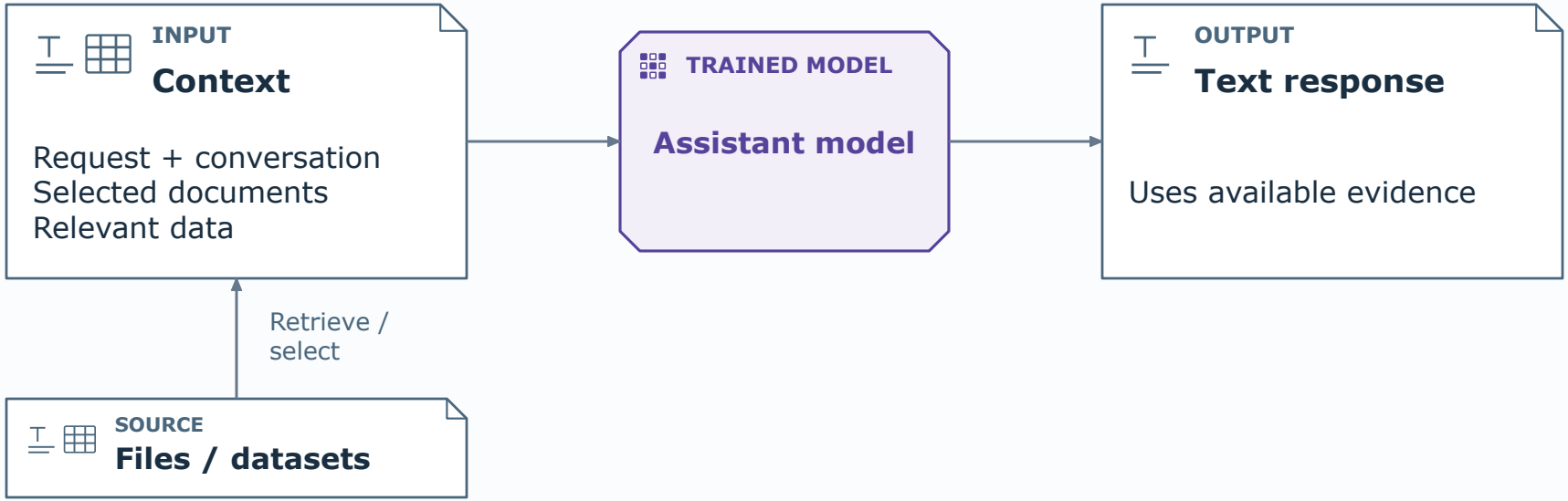
InstructGPT, early ChatGPT (2022 →)





Context and retrieval

Retrieval-augmented generation built around LLMs (RAG 2020, early ideas 2010s)



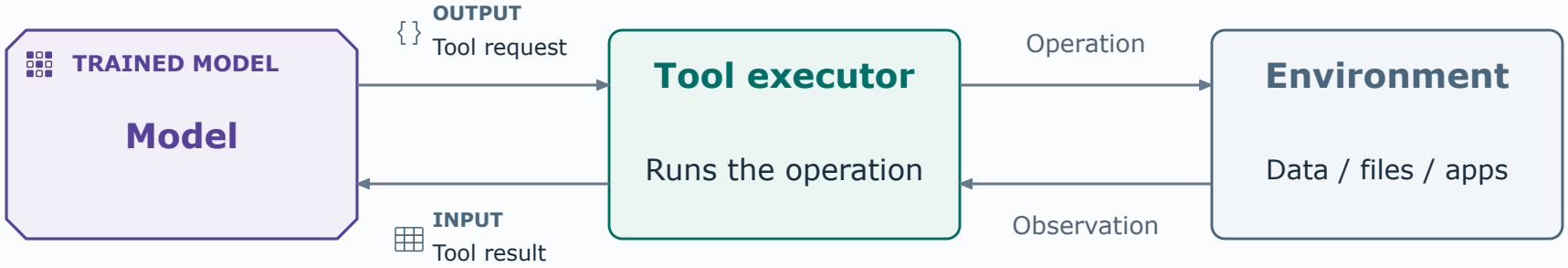
Multimodal input

GPT-4 with vision / GPT-4o (2023, earlier models 2014+)



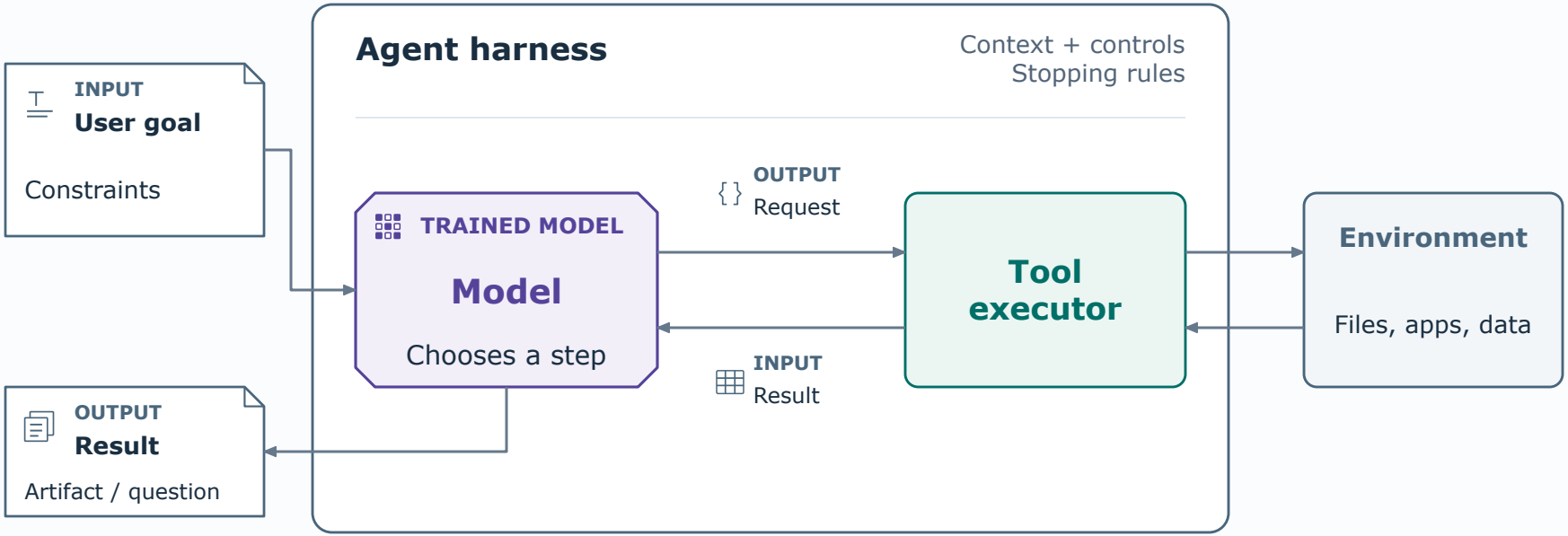
Tool requests and execution

GPT-3.5 / GPT-4 function calling and tool-enabled assistants (2023, early tool-calling 2021+)



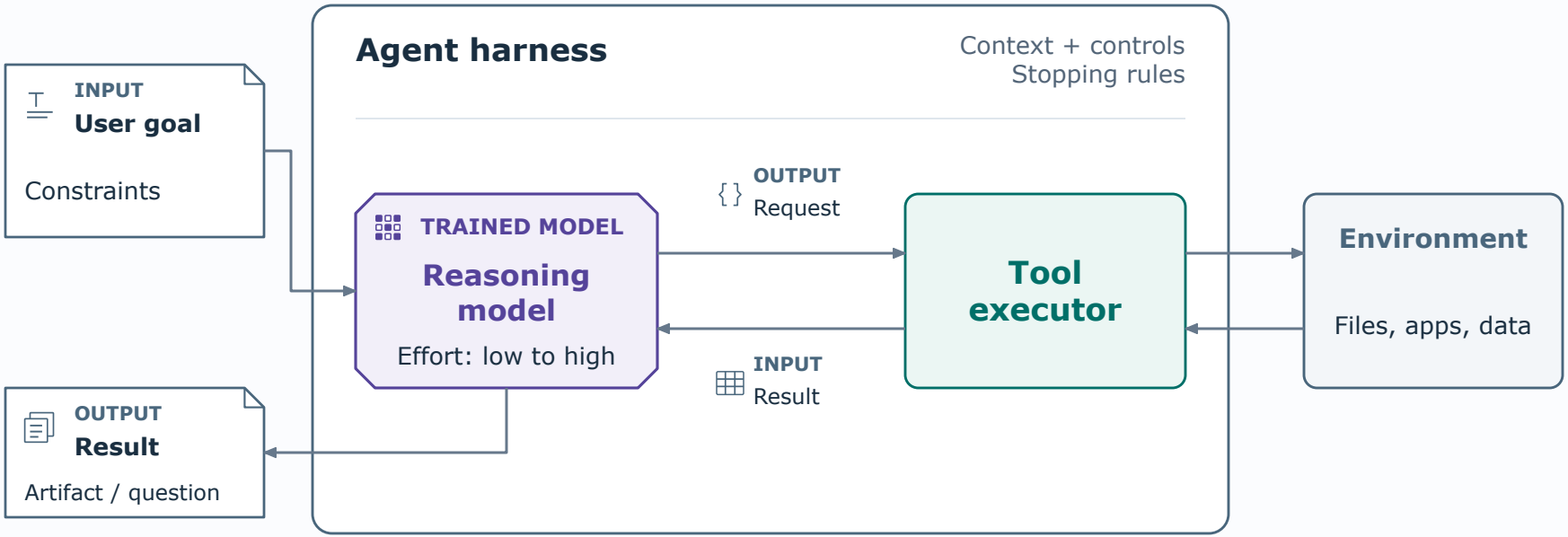
The agent feedback loop

Common architecture for LLM agents



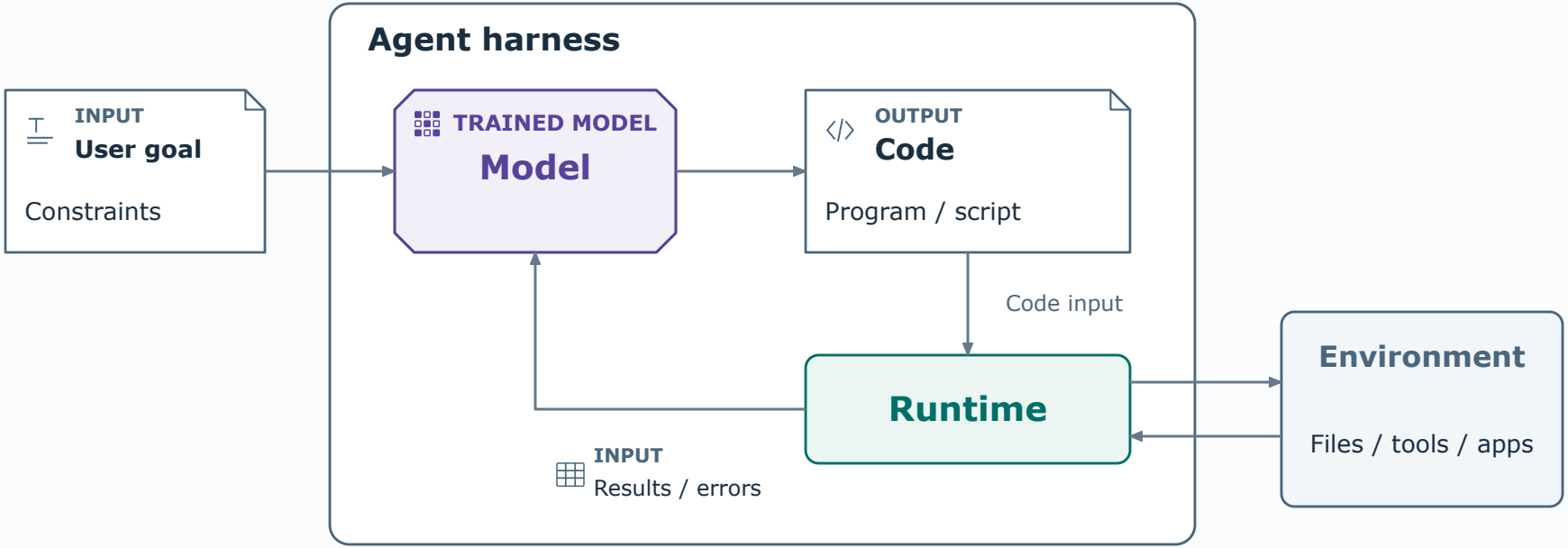
Reasoning effort and agency

"Show your work" 2021, "Chain-of-thought Prompting" 2022, OpenAI's o1 2024



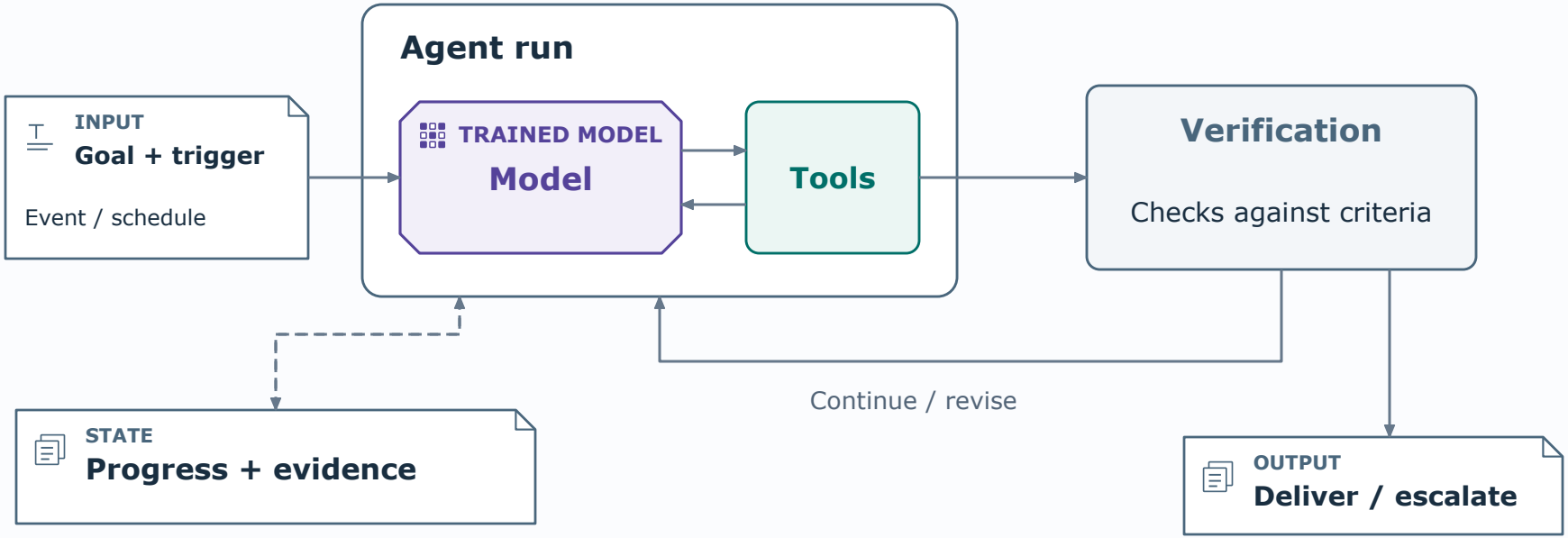
Code as an action interface

Code as Policies (2022), CodeAct (2024), …



Engineering loop

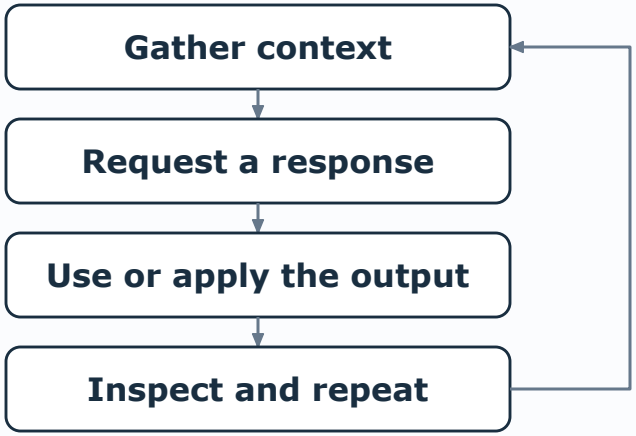
Possibly also orchestration across agent runs



From user coordination to agents

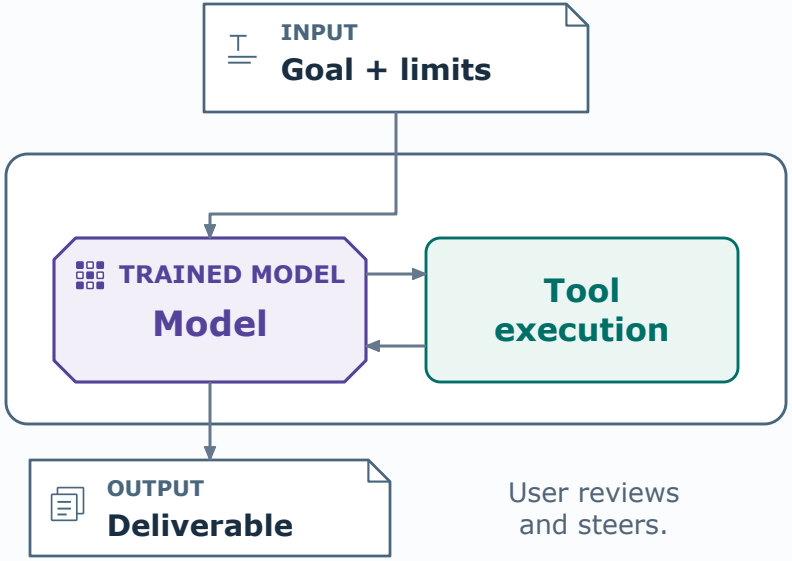


User coordination



User reviews and chooses.

Agent coordinates more steps

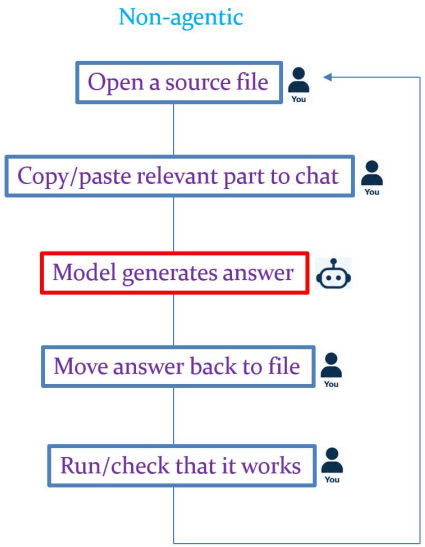


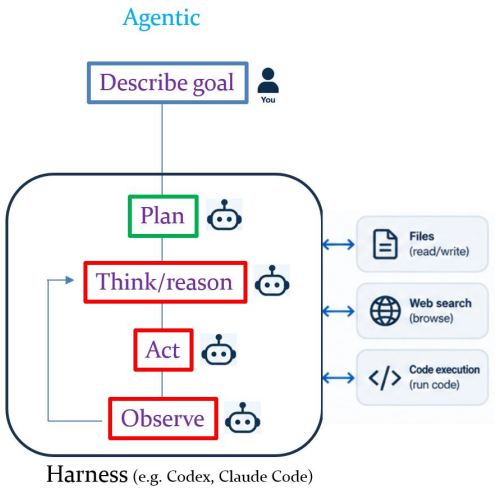
User reviews and steers.



The starting point was the following diagram and a prompt for a quick overview of evolution of LLMs to current agents, in about 10 minutes.

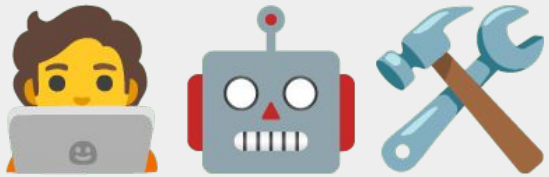
The previous slides were co-created with agents; more about creating slides is discussed later.





The rest was refining and fact-checking

AI Agents in Software Engineering



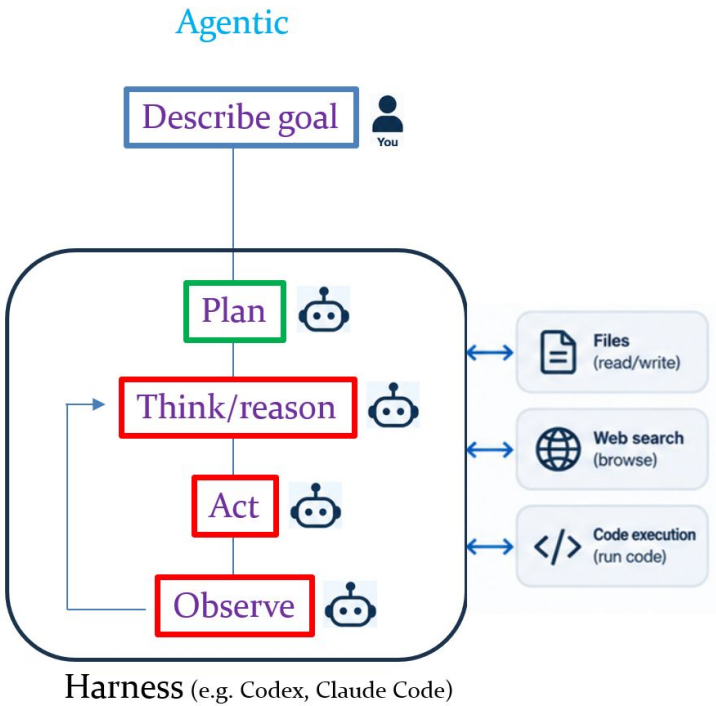


AI Agents in Software Engineering

Contemporary AI agents can carry a large proportion of development tasks.

- Implement features
- Fix bugs
- Write tests
- Refactor code.
- …

The developer's role is primarily defining goals and constraints, making design decisions, and validating outcomes.



Agents use "skills", which are effectively prompts that guide the agent what to do.



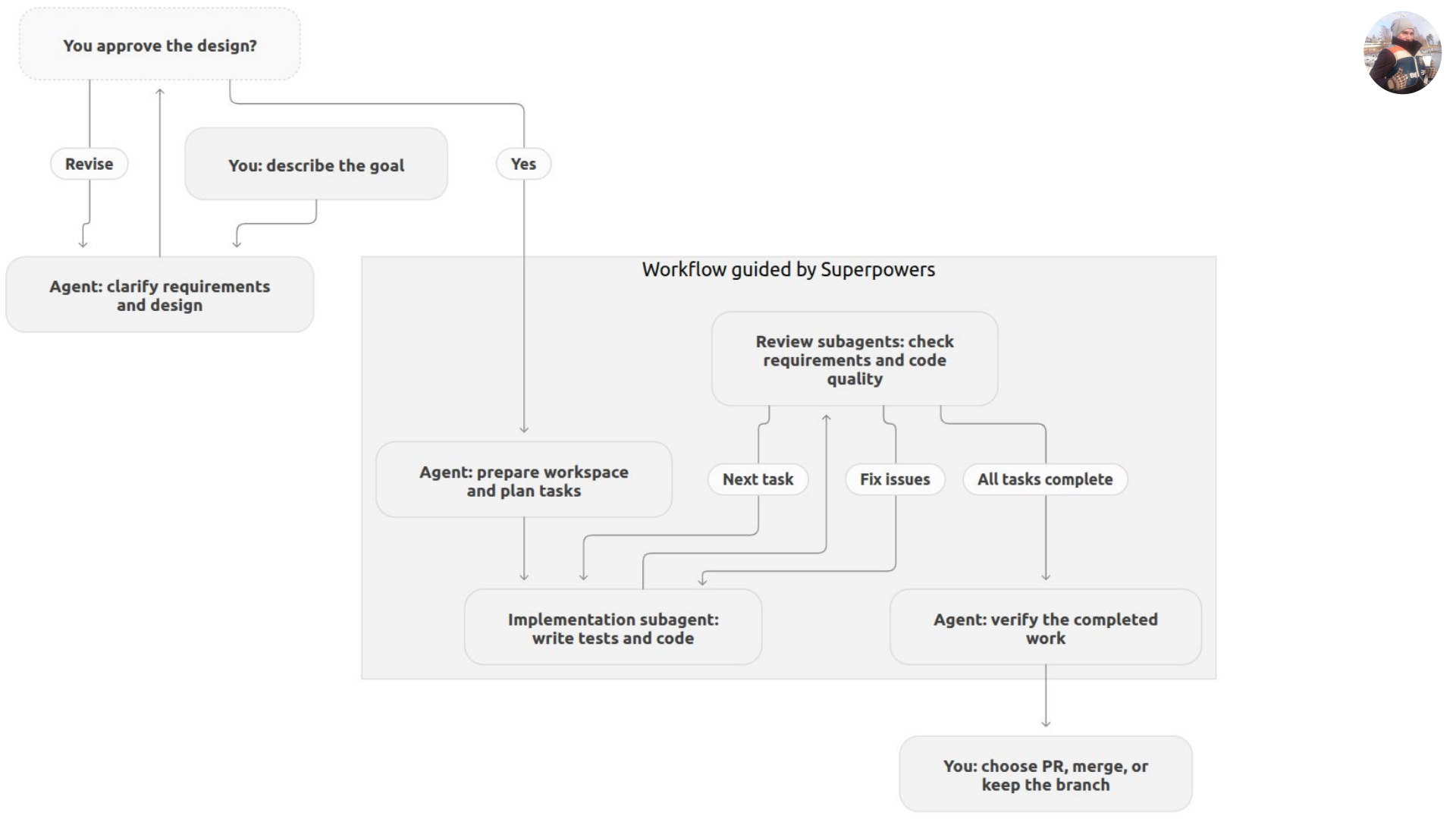
Skill libraries (workflow layers) like “superpowers” give coding agents a structured way to work.

Structure = task-to-skill mapping, where skills are prompts.

Possible set of skills:

1. **brainstorming**
2. **using-git-worktrees**
3. **writing-plans**
4. **subagent-driven-development** or **executing-plans**
5. **test-driven-development**
6. **requesting-code-review**
7. **finishing-a-development-branch**

https://github.com/obra/superpowers


You approve the design?
Revise
You: describe the goal
Yes
Agent: clarify requirements and design
Workflow guided by Superpowers
Review subagents: check requirements and code quality
Agent: prepare workspace and plan tasks
Next task
Fix issues
All tasks complete
Implementation subagent: write tests and code
Agent: verify the completed work
You: choose PR, merge, or keep the branch

Handle single-prerequisite download locks end to end #54

New issue

Could also work fully on GitHub...

Closed | Parent: Block downloads for locked assignments

manh-bui opened on Sep 11 — Member

Parent

#46

What to build

Deliver the first complete locked-download path for an assignment with one incomplete prerequisite. Preserve the platform's structured HTTP 403 response, normalize it as an assignment-domain lock, stop before installing files, and show a modal warning that can make the prerequisite current and reveal it in the selected course without downloading it automatically.

Acceptance criteria

- [ ] JSON API failures retain their parsed response body while preserving existing status, message, timeout, cancellation, and non-JSON fallback behavior.
- [ ] A validated `locked: true` HTTP 403 from either the starter metadata endpoint or starter-files endpoint becomes the same typed assignment-domain lock error.
- [ ] The normalized lock includes its reason, authoritative platform message, validated prerequisite exercise details, validated progress details when present, and optional platform code.
- [ ] Malformed lock payloads and unrelated API failures remain generic API errors.
- [ ] The obsolete `prerequisites_met` starter field, mocks, assertions, and service check are removed; HTTP responses are the sole lock authority.
- [ ] A download blocked by one incomplete prerequisite shows a modal **Assignment is locked** warning containing the platform message and a **Go to prerequisite** action

Assignees

No one - Assign yourself

Assign to Agent

Labels

agent-dispatched | agent-needs-review | ready-for-agent | repo:opencs-ide-plugin

Type

No type

Fields

Priority — Choose an option

Projects

Aalto OpenCS

Status — Done

Show 1 closed project

Milestone

No milestone

- Issues
- Assigned to me
- Created by me
- Mentioned
- Recent activity

- Views
- Projects
- Milestones
- Labels

Issues

New issue

is:issue state:closed label:ready-for-agent

Open 0 **Closed 16** Author Labels Projects Milestones Assignees Types Newest

- [x] Retry undelivered IDE action logs silently `agent-dispatched` `agent-needs-review` `enhancement` `ready-for-agent`
  aalto-opencs/opencs-ide-plugin#63 · by manh-bui was closed on Sep 14 — 1 — 1
- [x] Bound and compact active and completed activity storage `agent-dispatched` `agent-needs-review` `enhancement` `ready-for-agent`
  aalto-opencs/opencs-ide-plugin#62 · by manh-bui was closed on Sep 14 — 1 — 1
- [x] Send compact IDE action logs after successful submission `agent-dispatched` `agent-needs-review` `enhancement` `ready-for-agent`
  aalto-opencs/opencs-ide-plugin#61 · by manh-bui was closed on Sep 14 — 1 — 2
- [x] Rework IntroCS PR #805 to persist submission-linked IDE action logs `enhancement` `ready-for-agent`
  aalto-opencs/opencs-ide-plugin#60 · by manh-bui was closed on Sep 14
- [x] Guide students through multiple or unresolved prerequisites `agent-dispatched` `agent-needs-review` `ready-for-agent` `repo:opencs-ide-plugin`
  aalto-opencs/opencs-ide-plugin#56 · by manh-bui was closed on Sep 11 — 1
- [x] Apply structured lock handling to every lock reason and redownload `ready-for-agent` `repo:opencs-ide-plugin`
  aalto-opencs/opencs-ide-plugin#55 · by manh-bui was closed on Sep 11 — 1
- [x] Handle single-prerequisite download locks end to end `agent-dispatched` `agent-needs-review` `ready-for-agent` `repo:opencs-ide-plugin`
  aalto-opencs/opencs-ide-plugin#54 · by manh-bui was closed on Sep 11
- [x] Replace broad refreshes with targeted synchronization `ready-for-agent`
  aalto-opencs/opencs-ide-plugin#53 · by manh-bui was closed on Sep 10 — 1



Plenty of options to choose from, e.g. “Skills for Real Engineers” – ongoing development and debate on “what’s the best option”.

The state-of-the-art "out of the box" models (e.g. OpenAI GPT-6.1 Sol with Codex, Opus and Fable in Claude Code) already have built-in harnesses.

https://github.com/mattpocock/skills

How we (facilitators) use AI Agents



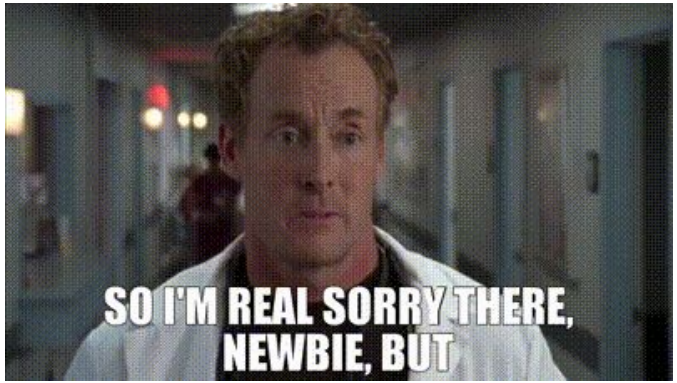


Paul’s magical PDF to ebook demo

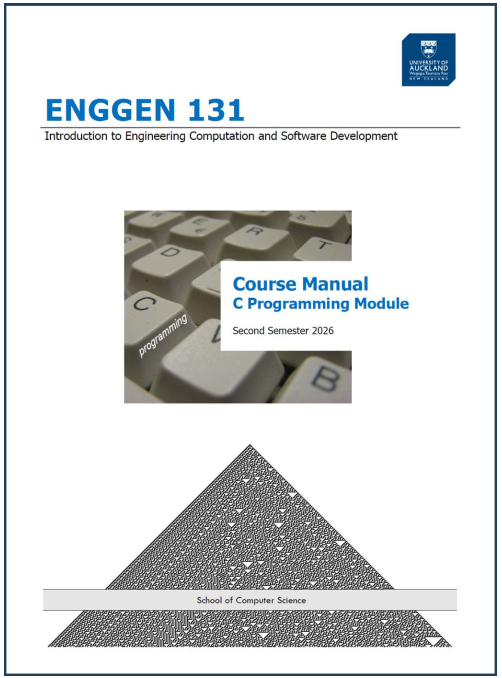


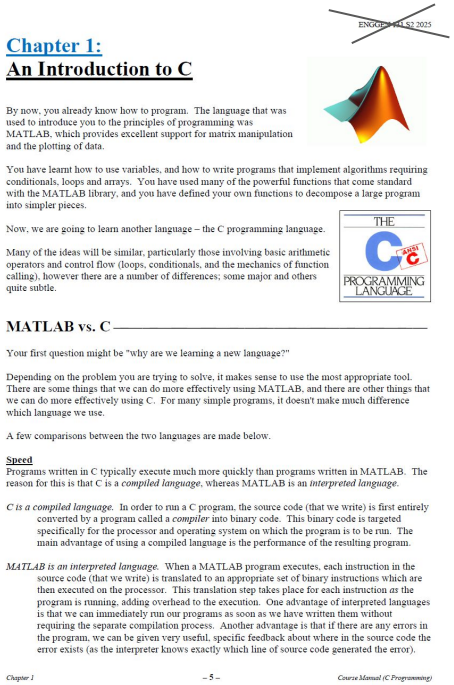
How we (facilitators) use AI Agents

Breathing new life into an archaic textbook



2026







Initial concern: permissions and file access

lder

Logged in with API key

Codex settings

Keyboard shortcuts

```
sandbox_mode = "workspace-write"
approval_policy = "on-request"

[windows]
sandbox = "elevated"
```

Terminal

Do you want to allow npm to download project-local dependencies from the registry?

```
npm install vite pdfjs-dist
```

Deny Esc Allow once

I couldn't make the edit because `Paul.txt` is outside the writable workspace, and the permission request to modify it was rejected.

No changes were made. The intended comment would be:

C

```c
/* Return the sum of two integers. */
int sum(int a, int b)
{
    return a + b;
}
```



A simplistic prompt

- That I wrote myself…

I have an old programming textbook as a PDF. I want to turn it into a modern, attractive and interactive website that students could use instead of reading the PDF.

Please inspect the textbook, understand its structure and content, and build the website for me. Preserve the important teaching content, but rethink how it is presented for the web. Add useful interactivity where it genuinely improves the learning experience.

You may create whatever files and project-local dependencies you need, but work only inside the current workspace and do not install system-wide software.

Before you finish, run the site locally, check that it works, and fix any obvious problems you find.



You get what you ask for

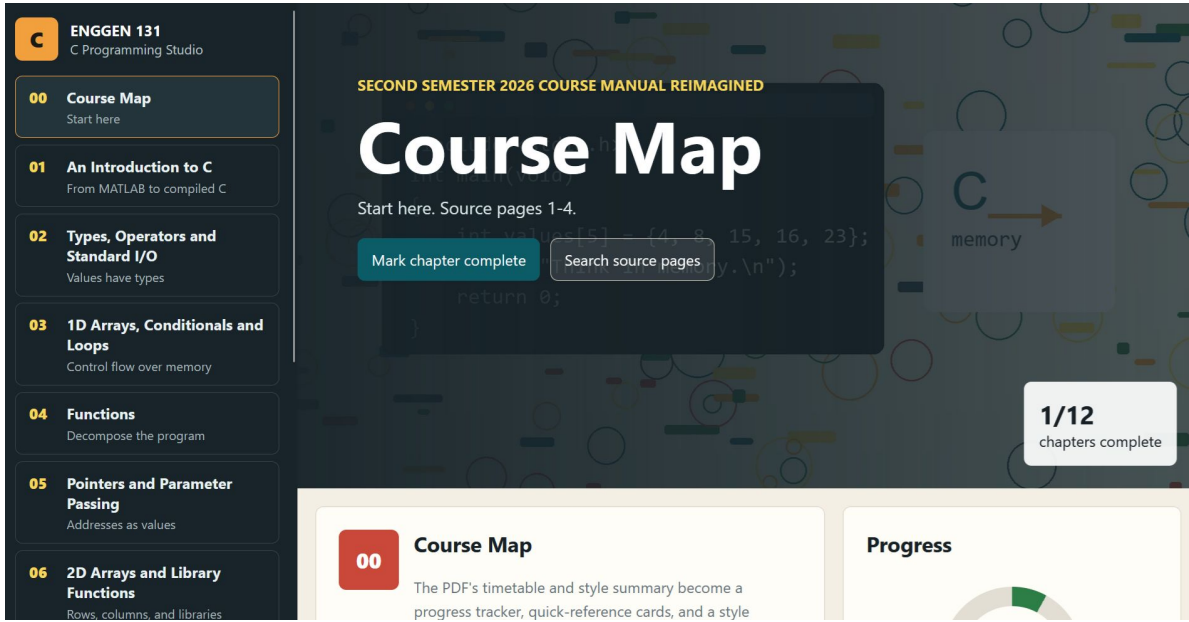
- Vite website; content poorly accessible in JS files; looks like something I would make:

Worked for 22m 39s >

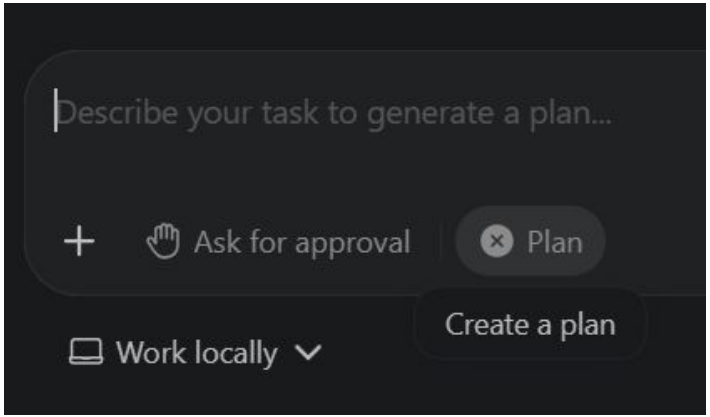
Built the interactive course website and left it running locally:

`http://127.0.0.1:5173/`

```
export const chapter4 = {
  blocks: [
    h("function-contracts", "Function contracts"),
    goals([
      "Define functions with a name, parameters, return type, and body.",
      "Call functions and use returned values.",
      "Declare prototypes before calls when needed.",
      "Trace execution across several function calls.",
      "Explain local scope and avoid accidental global state.",
    ]),
    p("A function packages a task behind a small contract: what inputs it n
    to call it."),
    p("Functions reduce overload because you can think about one small job
    function body contains the details of how the job is done."),
    code("A small function", "int MaxOfThree(int a, int b, int c)\n{\n    i
    = c;\n    return max;\n}"),
    list("Read a function header as a contract", [
      "int is the return type.",
      "MaxOfThree is the function name.",
      "int a, int b, int c are parameters.",
      "The function body does the work and returns one int value.",
    ]),
    p("Parameters are local names used inside the function. When main calls
    starts with 9, and c starts with 2 for that one call."),
    classify("ch4-header-match", "Checkpoint", "Match function headers to j
      { id: "print", label: "Prints but returns nothing" },
```



A bit of planning and UI design



I'll stay in Plan Mode and use the repo plus `Coursebook.pdf` as the grounding source. I'm going to inspect the site structure, current content generation path, and PDF-extraction options first, then I'll come back with any product decisions that genuinely need your input.

Retain, Revise, Replace

- Retain:
  - Vite setup.
  - Overall single-page app architecture.
  - Sidebar chapter navigation.
  - Progress tracking.
  - Source-page search concept.
  - Existing interactive widget idea.
- Revise:
  - Rendering components to support a real textbook block model.
  - Search to report both PDF page and printed coursebook
  - Widgets so they attach to exact source sections instead
  - themes.
  - CSS toward a denser coursebook reader layout.
- Replace:
  - Most of `src/courseContent.js`.
  - Current exercise summaries.
  - Current "Core Ideas" cards as the primary teaching surf
  - Any paraphrased chapter content that stands in for act

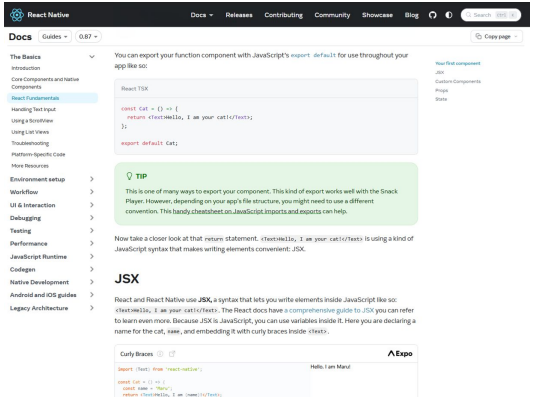
Assumptions

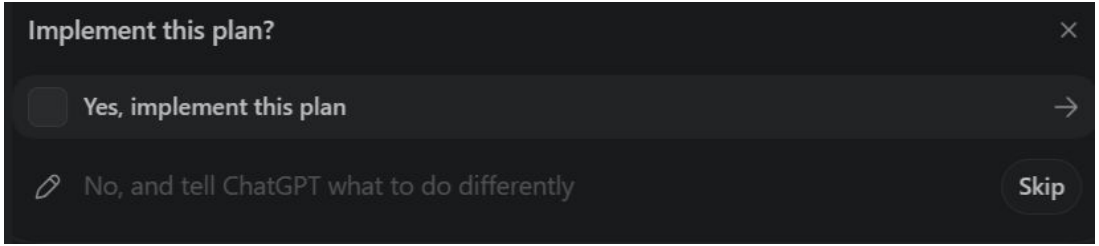
- The goal is complete faithful coverage of the C module PDF
  guide.
- Verbatim reproduction is acceptable because `Coursebook.p`
  authoritative source for this project.
- Chapter 8 should remain minimal because the source itself i
  Discussion" plus notes/visual material, not a fabricated proje
- The implementer should prioritize source fidelity over addin

I have added three visual reference files to the project root:

- `Example_UI_Layout_React1.png`
- `Example_UI_Layout_React2.png`
- `Example_UI_Layout_React3.png`

Please inspect all three images and use them as design inspiration.

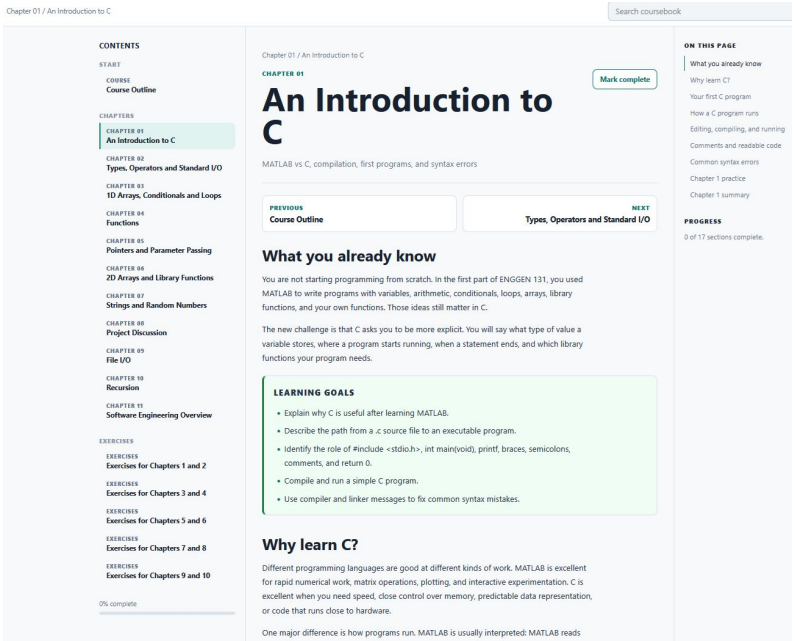


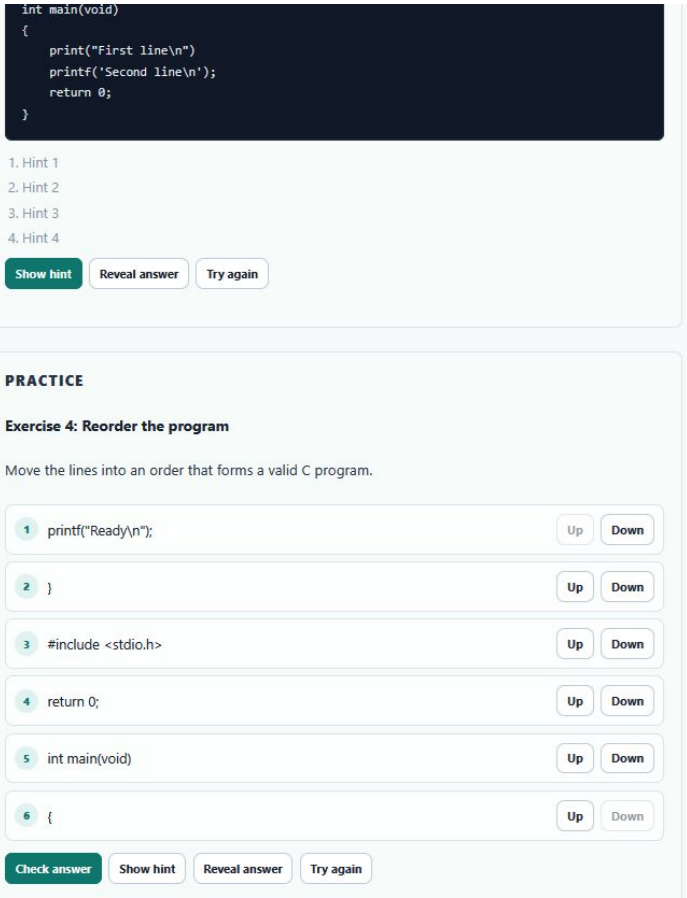




Iteration and improvement

- But not right technology choice…

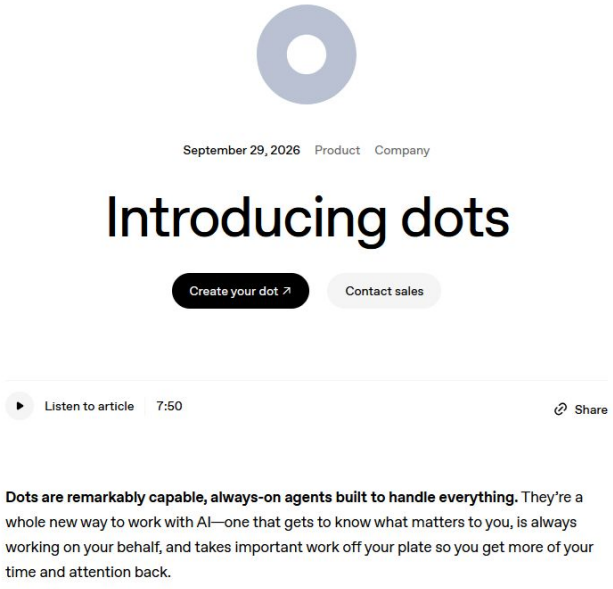


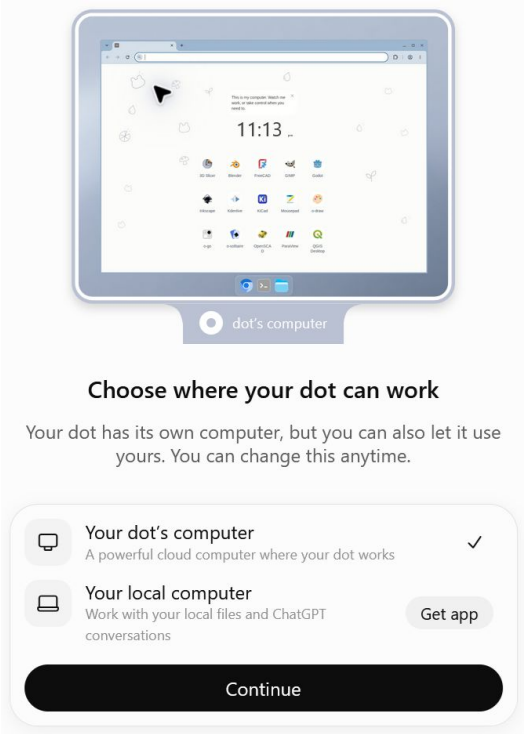




Next: do it all in the cloud

- “Dots”







Next: do it all in the cloud

- “Dots”

C_Coursebook_2026.pdf

I want to build an interactive website for an introductory programming course in C. I have uploaded a PDF of the coursebook that I wrote, but this is now old and static. The content is a good starting point though, but it could be improved and included additional detail. Please convert this to a high quality website that is interacted. I want it to be a very high quality and effective educational resource.

I'll turn your coursebook into an interactive C course, with clearer explanations, worked examples and exercises that help students check their understanding. It will take a few minutes, and I'll send the link when it's ready!

Your interactive C course is ready to review. It's private to you; you can ask me to change its sharing settings.

Automated content and interaction checks passed. Edited C code needs a separate compiler to run. Visual browser testing remains unverified because of the private sign-in.

I've set up the guest credentials now on the website.

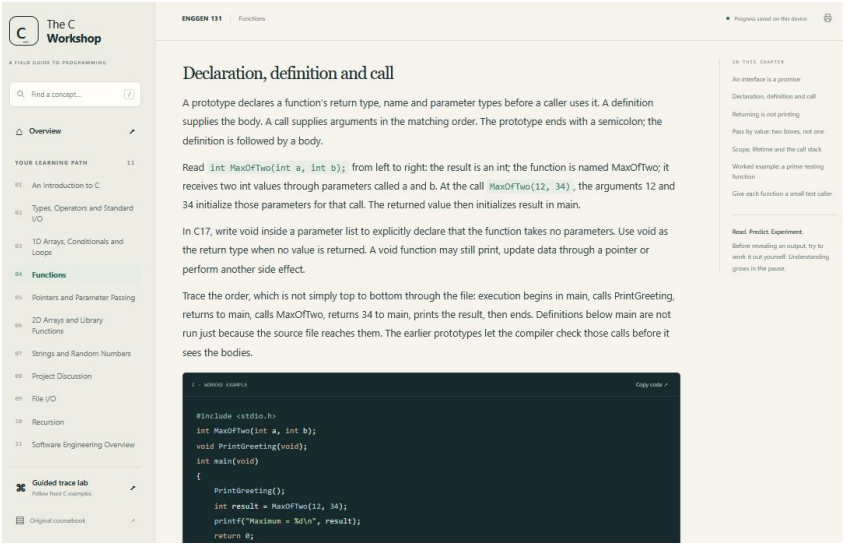
I think the next thing to do, before I release this widely, is to imagine whether a student who is a novice would actually find it useful. Can you please simulate a novice learner actually using the book, moving from one chapter to the next and completing all of the activities. If you feel that any of the explanations require improvement, please do that. I want the book to be very high quality and useful as a learning resource. Please be critical of the book's content, and make improvements were needed.

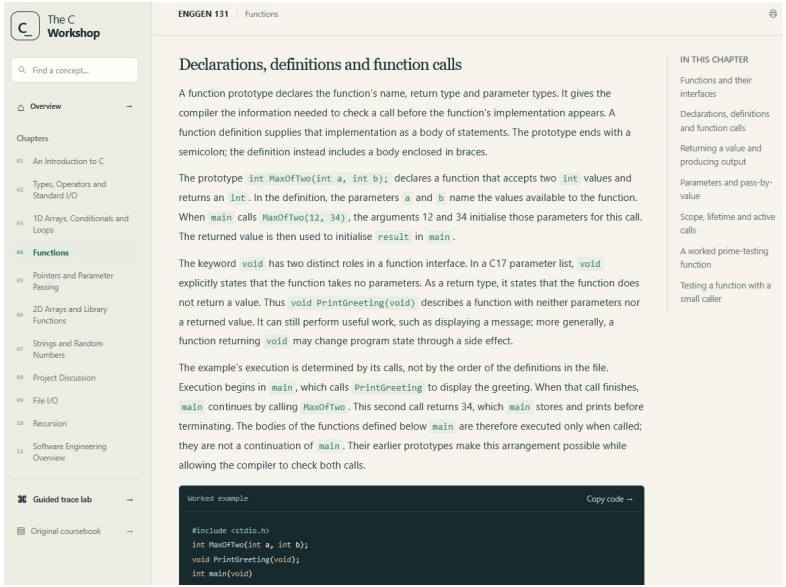
I'll work through it chapter by chapter as a novice, attempt every activity, and improve explanations wherever the learning journey breaks down. I'll also check the guest setup, and keep you posted as the review progresses. This will be a simulated learner review rather than evidence from real students.



Next: do it all in the cloud

- “Dots”

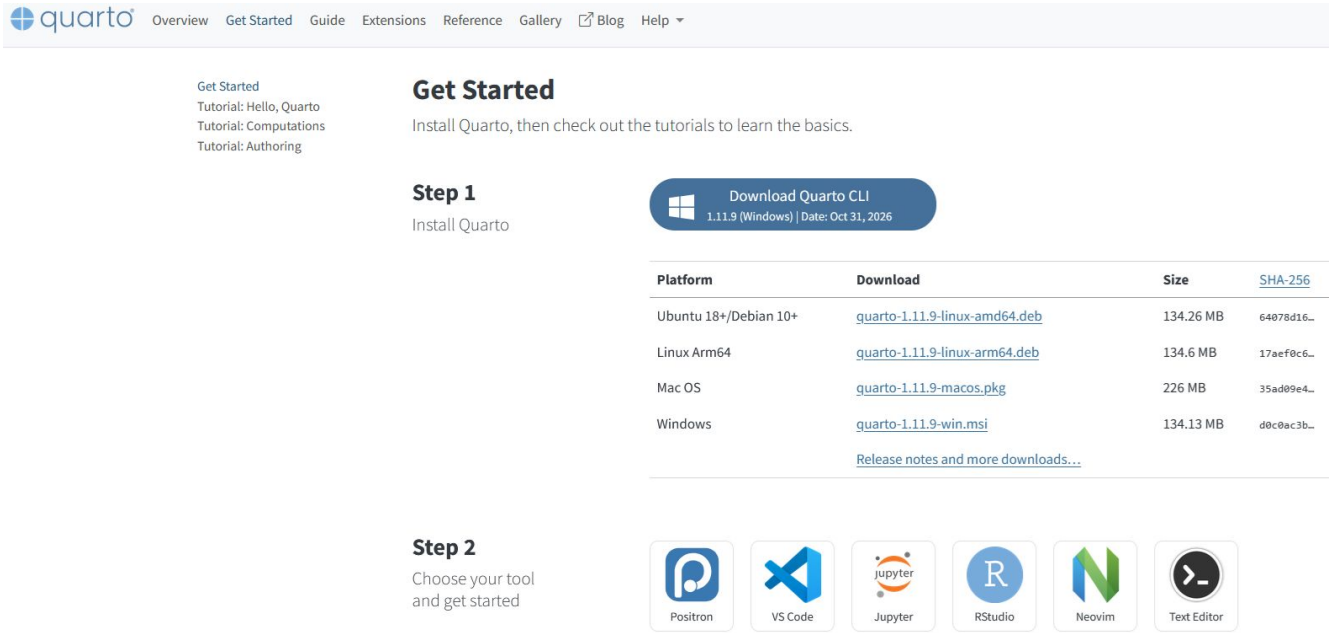






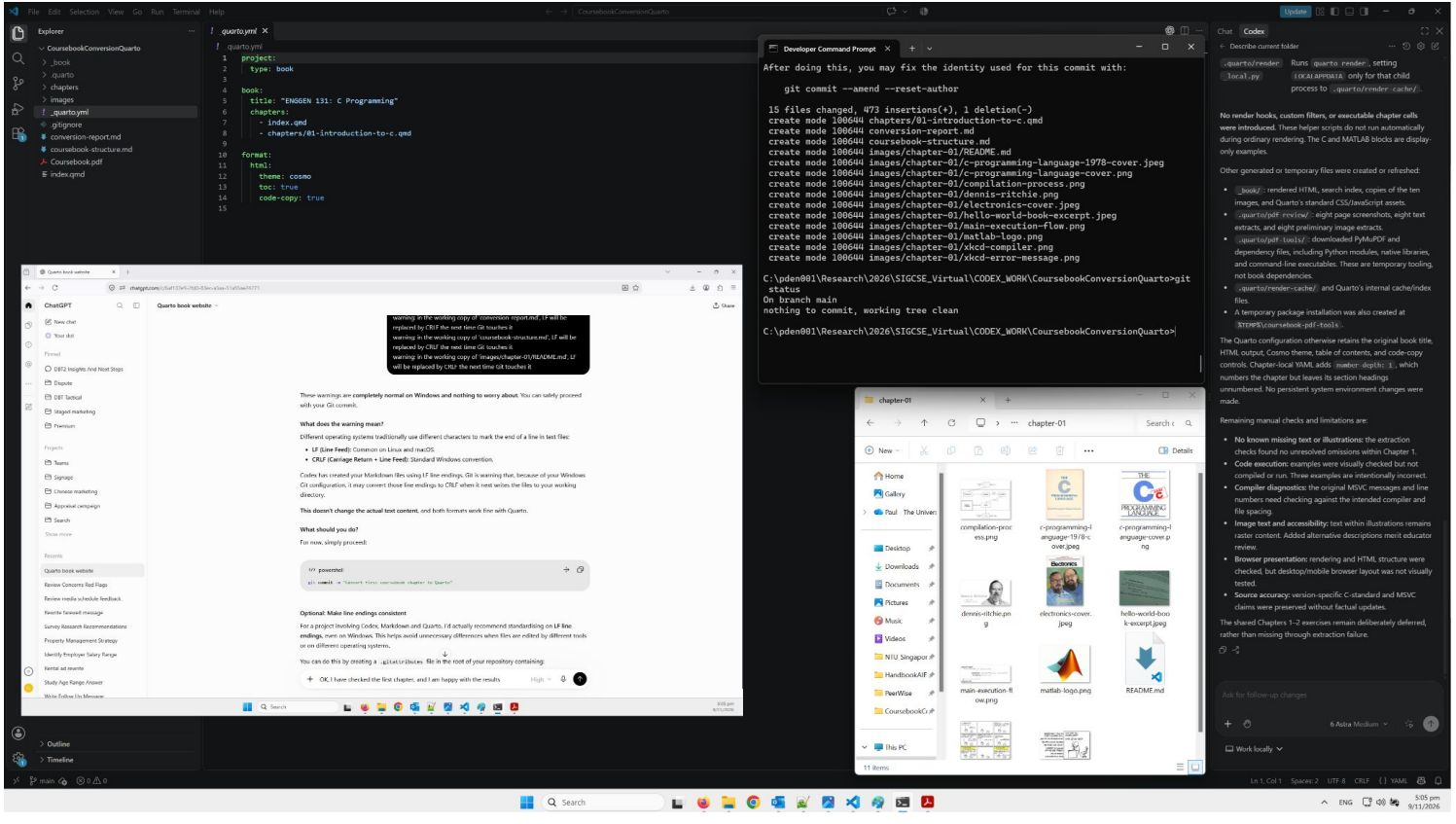
Next: try Quarto

- Well suited for the textbook conversion task (content – presentation)





ChatGPT for prompts





ChatGPT for prompts

I have created a Quarto Book project to convert an existing PDF coursebook into a maintainable, interactive educational website.
The source is `Coursebook.pdf`, located in the root directory.
My primary objective is to maintain the coursebook content as simple, human-editable Quarto Markdown (`.qmd`) files.
Please do the following:

1. Examine the PDF and identify its chapter structure, including chapter titles and main sections.
2. Create a `chapters/` directory and convert the first substantive chapter into a `.qmd` file. This first chapter should be the one called "Chapter 1: An Introduction to C"
3. Preserve the original educational content as faithfully as possible, including:
   - Explanatory text
   - Headings and subheadings
   - C programming examples, properly formatted as fenced code blocks
   - Tables
   - Figures and diagrams
   - Exercises and questions
4. Extract illustrations into an `images/` directory where practical, and reference them from the Markdown.
5. Update `_quarto.yml` to include the new chapter in the book navigation.
6. Keep the actual chapter content in Markdown rather than embedding it in JavaScript, HTML templates or React components.
7. Do not introduce Node.js, Vite, React or other website frameworks. Use Quarto's built-in formatting and navigation.
8. Do not paraphrase, summarise or invent coursebook material during conversion. Flag any content that cannot be extracted reliably rather than guessing.
9. Produce a short conversion report identifying any missing diagrams, uncertain content, formatting problems or code examples that need manual verification.
10. Run `quarto render` and resolve any rendering errors.
For this first iteration, focus on converting the content accurately into a readable Quarto Book. Do not introduce interactive exercises or other enhancements yet.

The finished book should be maintainable by an educator editing the `.qmd` files directly in a text editor.



Easier to maintain

ENGGEN 131: C Programming

Welcome
1 An Introduction to C
2 Types, Operators and Standard I/O
3 1D Arrays, Conditionals and Loops
4 Functions
5 Pointers and Parameter Passing
6 2D Arrays and Library Functions
7 Strings and Random Numbers
8 Project Discussion
9 File I/O
10 Recursion
11 Software Engineering Overview

```c
#include <stdio.h>

int SameWholeNumber(int x, double d);

int main(void)
{
     if ( SameWholeNumber(10, 10.9) ) {
         printf("same\n");
     }

     return 0;

}

int SameWholeNumber(int x, double d)
{
     int truncated = (int)d;
     return x == truncated;
}
```

function call

function definition

When the `SameWholeNumber()` function is called, the value 10 is assigned to the input parameter `x` and the value 10.9 is assigned to the input parameter `d`, before the function body executes.

In this case, the function would return 1 (or true, because the whole part of the `double` value 10.9 is 10) and so the output of the program would be:

```
same
```

Notice that in the function call, the input values are separated by a comma. Also, the order is

Table of contents
- Learning outcomes
- MATLAB Comparison
- Functions
  - Defining a function
  - Identifier names
  - void
  - return
  - Returning 0 from main()
  - Calling a function
  - Prototypes
  - No parameters
  - Multiple parameters
- Variable scope
- Chapter 4 Summary Program

01-introduction-to-c.qmd

02-types-operators-and-standard-io.qmd

03-arrays-conditionals-and-loops.qmd

04-functions.qmd

05-pointers-and-parameter-passing.qmd

06-2d-arrays-and-library-functions.qmd

07-strings-and-random-numbers.qmd

08-project-discussion.qmd

09-file-io.qmd

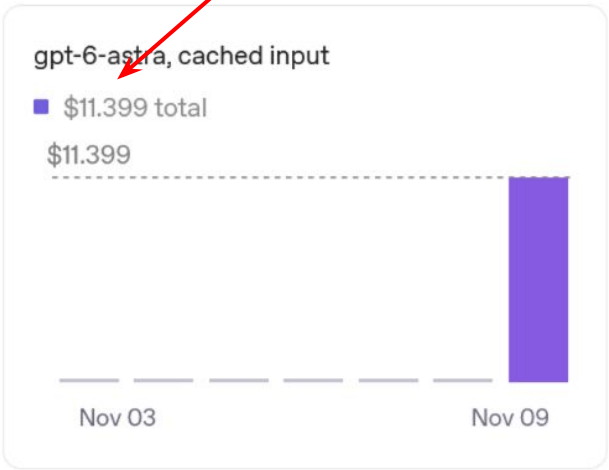
10-recursion.qmd

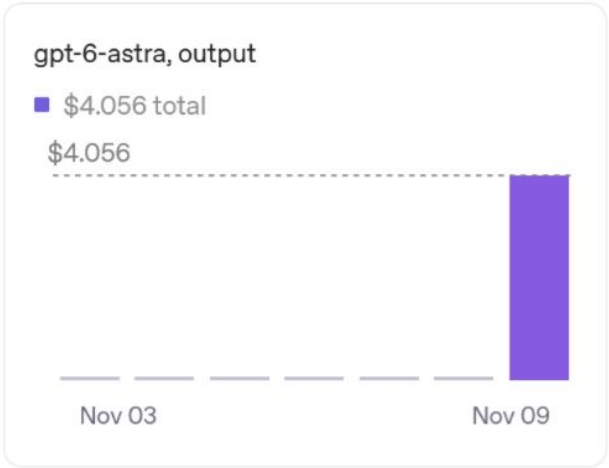
11-software-engineering-overview.qmd

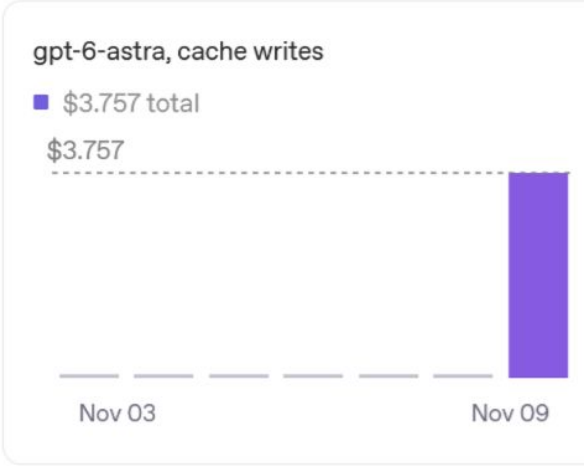


60 minutes; < $20 (GPT 6 Astra)

Not eleven thousand dollars and change, but eleven point change dollars.









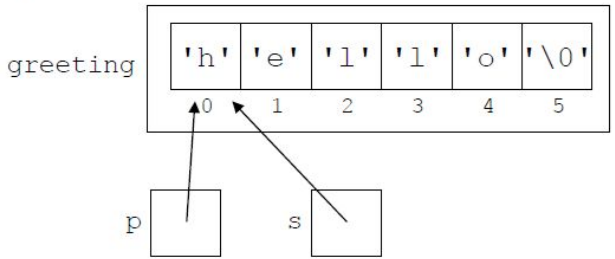
Maybe 2027 will be the year!

char *

An array variable behaves like a pointer to the first element in the array (we have seen this when we pass an array to a function as a parameter - a pointer to the first element of the array is passed to the function). For this reason, it is very common to use the type "pointer to `char`" when working with strings. Consider the following code:

```
char greeting[] = "hello";
char *p, *s;

p = greeting;
s = p;
```



In this case, both `p` and `s` point to the same string, so any changes made to the string through one pointer will also be seen by the other pointer. We can use either the pointer variables, or the array variable to access the string, so the following statements:

```
printf("%s", greeting);
printf("%s", p);
```

ENGGEN 131: C Programming

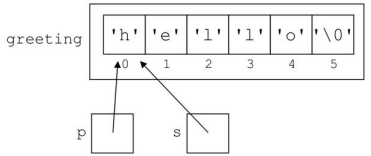
Welcome
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`char *`

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```
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printf("%s", p);
```



Oh, one more thing…



Geoff’s great idea…

“You might want to point the agent at WCAG 2.1 and see what it can do?”



Oh, one more thing…

REPORT CONTENTS

1. Assessment outcome
2. Standard and method
3. Confirmed findings
4. Additional issues and review items
5. Criterion-by-criterion coverage
6. Verification still needed
7. Evidence and reproducibility
8. Embedded evidence archive

Accessibility audit: ENGGEN 131 C Programming

Assessment date: 15 November 2026 (Pacific/Auckland)
Target: WCAG 2.1 Level AA, including Level A
Snapshot: Git commit `51896cc` — "First pass all chapters"
Scope: Existing local Quarto HTML book; assessment only

Assessment outc

The assessed book does r
table relationships, text co
announcements. This conc

Seven finding groups belo
elements, and failed criteri
in generated Quarto comp

The assessment combines

Confirmed findings

Priorities express user impact and reach, not normative WCAG severity. "High" identifies substanti
learning content or navigation; "Medium" identifies a material, narrower barrier. Recommendation
outcomes for a later remediation pass; none has been implemented.

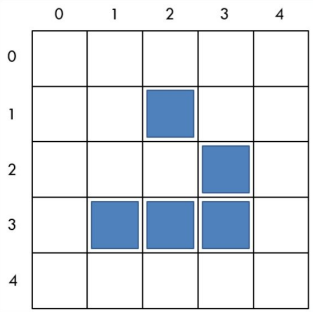
A01 — Meaningful images lack equivalent descriptions

High · SC 1.1.1 (A) · Confirmed examples; wider image inventory requires individual review.

All 178 images have `alt` attributes, so a missing-alt rule passes. However, 91 say only "Illustration
page …", and another 73 give a diagram title plus "reproduced from printed page …". These **164 review candidates are not an assertion of 164 failures**: some images repeat nearby text, and some diagram details may be incidental.

Confirmed examples:

Source	Missing information
`chapters/06-2d-arrays-and-library-functions.qmd:663`	The Game of Life starting grid has live cells at zero-based `(row, column)` positions `(1,2)`, `(2,3)`, `(3,1)`, `(3,2)`, `(3,3)`. The alt text only identifies printed page 97. Nearby prose gives the grid size and number of live cells, but not their positions, which are needed to follow the generations.
`chapters/08-product-discussion.qmd:15`	The comic depicts a machine-learning "pile of linear algebra"



A01 · The image conveys exact cell positions that its generic alternative and surrounding prose omit.



A01 · The Chapter 8 comic has meaningful dialogue, but the book's alternative only identifies a source page.



Arto’s Platform Development and Styling

Platform design system analysis

some sort of a "the exercises are visible on user interfaces larger than x"?)

Reading, Refining design ×10, Finishing up

Wide screens: the page shell now caps at 1760px with hairline edges, the reading measure stays at 68ch but centers in its column, the practice rail scales `clamp(420px, 30vw, 560px)`, and paddings/title size scale with viewport.

Below 1080px the rail is hidden and the reading column gains a "Practice — graded exercises open on wider screens" note stating the 1080px threshold.

Edited Skill Material Page.dc.html

Thanks. Maybe adjust the 5 more chapters to something like subtle arrows to the left and right instead? (the left arrow could be on the left before the currently visible one, while the right arrow could be on the right, where the three dots currently are).

Refining design ×3, Refining logic ×4, Finishing up

The "··· 5 more chapters" toggle is gone. The navigator now shows a two-chapter window with subtle ‹ / › chevrons on either side that step through all 7 chapters; each fades to light grey when there's nothing further that way.

Edited Skill Material Page.dc.html



Thank you. Now, let's focus on the normal material pages. I've attached four screenshots that show what the material pages look like with exercises and an opened python repl with and without the sidebar with

Describe what you want to create...

Opus 5.5 Medium

Send

Skills Page
13 pages

100%

Comment

Edit

Present

Share

A

learnerState returning

masteryTarget 80%

showUpNext

OpenCS / Skills / HTML

Anonymous User Login Register

SKILL

HTML

HTML is the language used to describe the content and structure of web pages. In this skill, you will learn how to write HTML documents, add text, links, images, lists, tables, semantic structure, and forms.

The focus is on clear, readable HTML that works in browsers, supports accessibility, and fits into contemporary web development with CSS, JavaScript, and component frameworks.



Last practised — 2 days ago · 2 of 10 topics at target

Continue

Quick practice

TOPICS MASTERED	OVERALL MASTERY	PRACTICE SESSIONS	TARGET
2/10	38%	14	80%

TARGET

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Send

Skills Page
13 pages

100%

Comment

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Share

learnerState returning

masteryTarget 80%

showUpNext

More formats and apps

Other

PNG Pick an element to export

Video MP4, for animation pages

Your destinations

Claude Code Send

Connect more destinations

OpenCS / Skills / HTML

SKILL

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HTML is the language used to describe the content and structure of web pages. In this skill, you will learn how to write HTML documents, add text, links, images, lists, tables, semantic structure, and forms.

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learnerState returning

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OpenCS / Skills / HTML

Anonymous User Login Register

SKILL

HTML



Continue

Quick practice

TOPICS MASTERED	OVERALL MASTERY	PRACTICE SESSIONS	TARGET
2/10	38%	14	80%

TARGET

Hand off to Claude Code

Local agent

Web session

Skills Page.dc.html

Add more files

Claude Code can read every file — it just focuses on these.

Add instructions (optional)

Open Claude Code and paste in this prompt.

Copy prompt

Platform design system analysis

some sort of a `the exercises are visible on user interfaces larger than x"?)

Reading, Refining design ×10, Finishing up

Wide screens: the page shell now caps at 1760px with hairline edges, the reading measure stays at 68ch but centers in its column, the practice rail scales `clamp(42 x, 30vw, 560px)`, and paddings/title size scale with viewport.

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Describe what you want to create...

Opus 5.5 Medium

Send

Skills Page
13 pages

100%

Comment

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Present

Share

A

learnerState returning

masteryTarget 80%

showUpNext

OpenCS / Skills / HTML

Anonymous User Login Register

Hand off to Claude Code

Local agent | Web session

Skills Page.dc.html

Change files

Claude Code can read every file — it just focuses on these.

Add instructions (optional)

Copied

Copied — paste it into your agent

```
Use the claude_design MCP (https://api.anthropic.com/v1/design/mcp, auth via
/design-login) to import this project:
https://claude.ai/design/p/2a945664-991d-458e-a4a8-1316e70a0035?
file=Skills+Page.dc.html

Focus on these files (the whole project is readable):
- `Skills Page.dc.html`

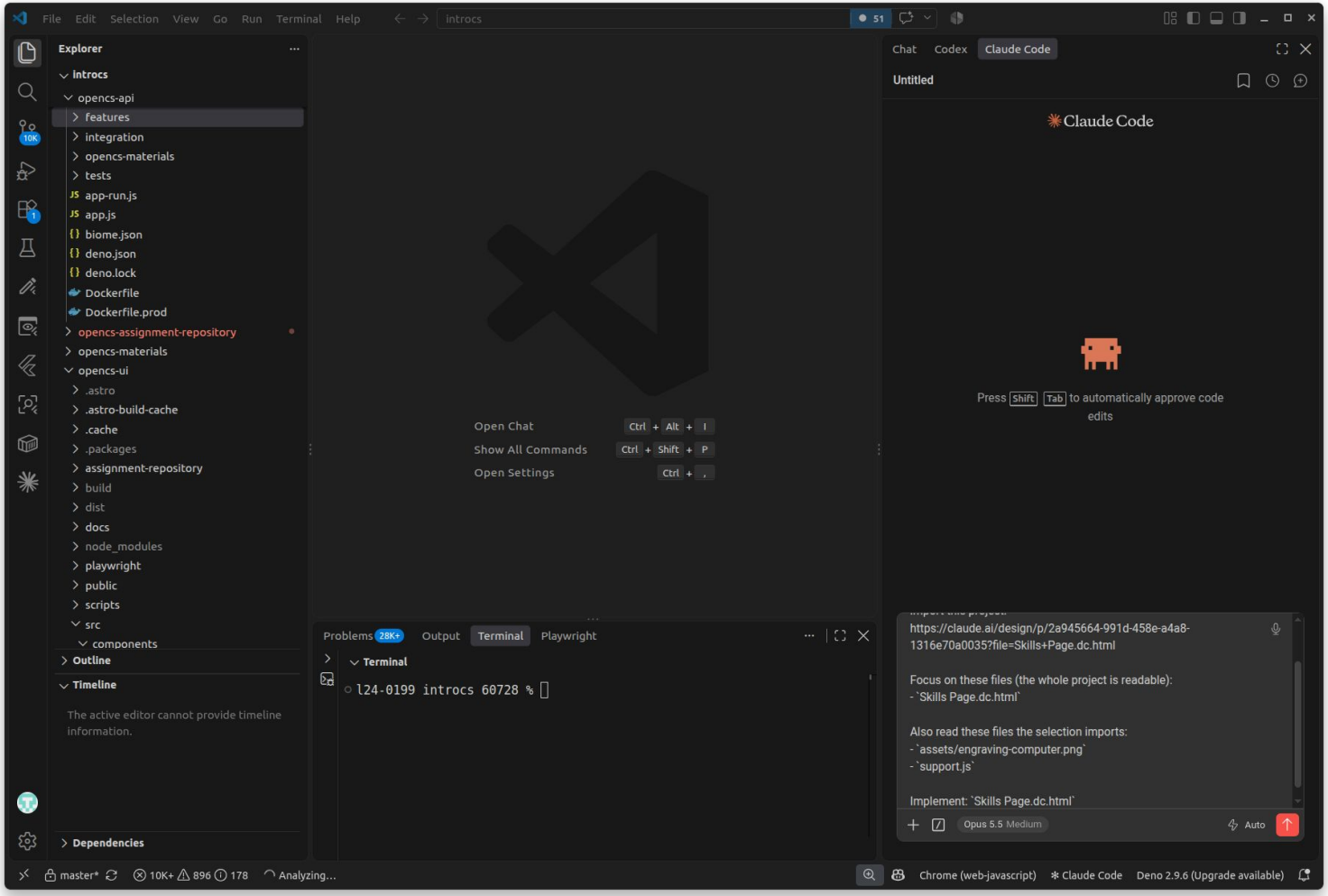
Also read these files the selection imports:
- `assets/engraving-computer.png`
- `support.js`

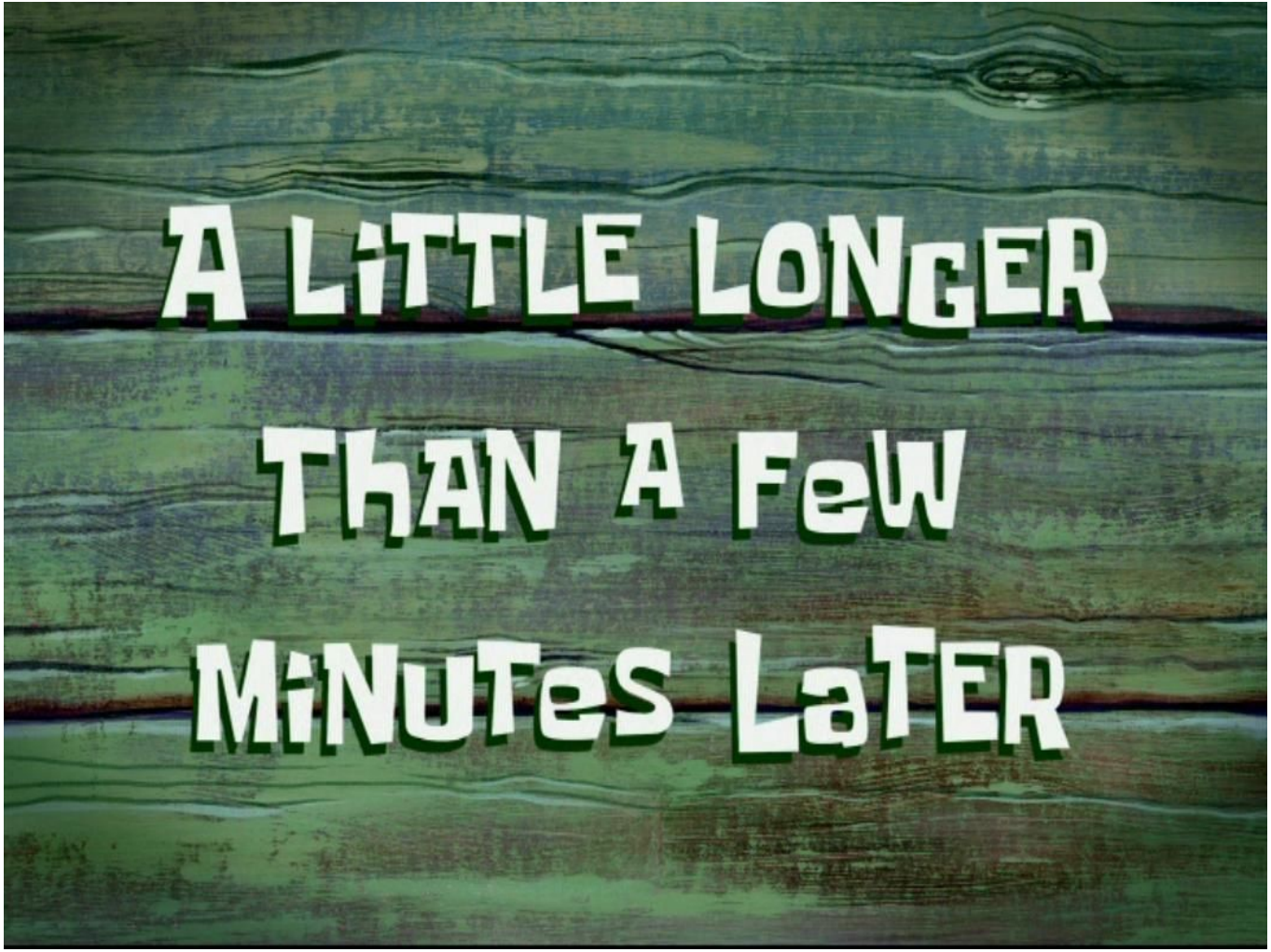
Implement: `Skills Page.dc.html`
```



TOPICS MASTERED	OVERALL MASTERY	PRACTICE SESSIONS	TARGET
2/10	38%	14	80%

TARGET


File Edit Selection View Go Run Terminal Help
introcs
Explorer
introcs
opencs-api
features
integration
opencs-materials
tests
app-run.js
app.js
biome.json
deno.json
deno.lock
Dockerfile
Dockerfile.prod
opencs-assignment-repository
opencs-materials
opencs-ui
.astro
.astro-build-cache
.cache
.packages
assignment-repository
build
dist
docs
node_modules
playwright
public
scripts
src
components
Outline
Timeline
The active editor cannot provide timeline information.
Dependencies
Open Chat Ctrl + Alt + I
Show All Commands Ctrl + Shift + P
Open Settings Ctrl + ,
Problems 28K+
Output
Terminal
Playwright
Terminal
l24-0199 introcs 60728 %
Chat
Codex
Claude Code
Untitled
Claude Code
Press Shift Tab to automatically approve code edits
https://claude.ai/design/p/2a945664-991d-458e-a4a8-1316e70a0035?file=Skills+Page.dc.html
Focus on these files (the whole project is readable):
- `Skills Page.dc.html`
Also read these files the selection imports:
- `assets/engraving-computer.png`
- `support.js`
Implement: `Skills Page.dc.html`
Opus 5.5 Medium
Auto
master*
10K+ 896 178
Analyzing...
Chrome (web-javascript)
Claude Code
Deno 2.9.6 (Upgrade available)



Including a bit of prompting and reviewing to get things fit the existing architecture...

SKILL

Python

Python is a general-purpose programming language used for automation, data processing, web services, command-line tools, testing, scientific computing, and everyday software work.

Last practised — never. Start anywhere; the first topic assumes nothing.

Start now | Quick practice



TOPICS MASTERED	OVERALL MASTERY	IN PROGRESS	TARGET
0/30	0%	0	65%

TARGET

UP NEXT



Exercise Generation and Validation

Classic approach with prompting -
now a lot easier with agents.

Automatic Generation of Programming Exercises and Code Explanations Using Large Language Models

Sami Sarsa, Paul Denny, Arto Hellas, Juho Leinonen

Aalto University
School of Science

THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

bit.ly/icer2022

Analyze the introductory python course materials at [path] and the exercise shape at [path].

Create a plan for an exercise sequence of five exercises for the chapter classes and games in part 6.

Output:

I recommend five exercise cards: three programming exercises and two short review sequences, all using the existing Asteroids game. Split the larger assignment so students finish a playable game before tackling persistence.

The course consistently asks students to predict, implement, test, and explain behavior inside supplied graphical programs. By this chapter, students have already studied composition, object identity, modules, independent records, and reconstruction. The sequence should apply those ideas while keeping the graphics, geometry, validation, and file operations supplied.

The chapter currently has four cards worth 73 points. Its 35-point Asteroids assignment already separates firing, collisions, snapshots, and restoration into grading stages. Splitting it at the existing 20-point gameplay / 15-point persistence boundary gives five cards without increasing the workload. See the [current assignment stages (line 29)]([path]/assignment-details.yml:29).

Order	Proposed exercise	Format	Points	Main outcome
1	Model moving objects	Programming	20	Give individual objects independent behavior
2	Predict collisions and responsibilities	Review sequence	10	Trace interactions before implementing them
3	Coordinate and play Asteroids	Programming	20	Coordinate objects through composition

…

Implement the plan

(or, iterate on the plan, giving feedback until satisfied, and then ask the agent to implement the plan...)

Output:

Implemented the [five-exercise sequence]([path]), preserving 73 points.

Gameplay and persistence now have separate starters, solutions, sample checks, and graders. Reviews cover collision predictions and model verification.

Validation passed: all reference graders and sample checks, partial-credit scenarios, defective submissions, quiz examples, MDX compilation, and exercise references.

Thank you.

Now, as ten students of varying competence levels, starting from novice to a student who is very good, create a range of correct and incorrect solutions to the exercises in the exercise sequence.

For each candidate solution, run the tests against the solution, and verify that the feedback from the tests is not misleading and that the feedback is such that it helps the student proceed.

Fix any identified issues.

Output:

I'll simulate ten students across a range of skill levels, run their candidate solutions through the graders, and review the feedback. I'll fix cases where the score or message could mislead a student or leave them without a useful next step.

… (a while later):

Created and tested 50 exercise attempts from ten simulated students, including 30 programming solutions.

- All 9 correct implementations pass.

- All 21 faulty implementations are rejected, with expected partial credit.

- Verified 90 quiz answers and 72 additional selection combinations.

Fixed a false rejection of valid constructors, misleading failures from unrelated unfinished methods, unclear error messages, and overly broad sample-success messages. Added boundary and restoration checks. No unhandled grading errors remain in the audited cases.



Slide Review, Generation and Enhancement

Aalto University

UNIVERSITY OF TORONTO



AI-Generated Slides: Are They Good? Can Students Tell?

Juho Leinonen*, Lisa Zhang*, Arto Hellas







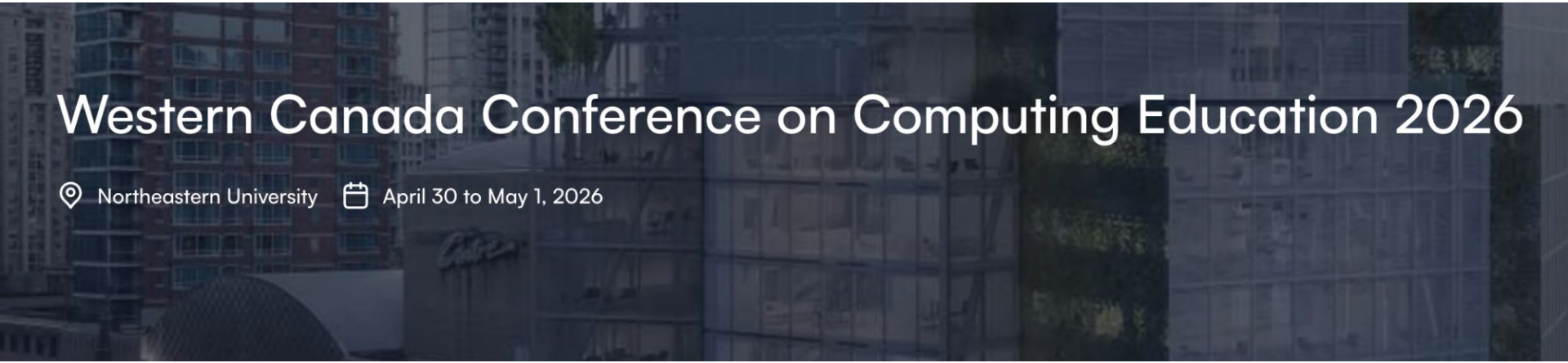
* Equal contribution

Aalto University

UNIVERSITY OF TORONTO



AI-Generated Slides: Are They Good? Can Students Tell?





Evaluation Plans

Arto





Evaluation Plans

Use Quarto to generate the slides. Include all makefiles, etc., so that the slides can be built. Please put the results in the <folder>

Arto





Evaluation Plans

Use Quarto to generate the slides. Include all makefiles, etc., so that the slides can be built. Please put the results in the <folder>

Arto vs Quarto







2 hour lecture

Lecture Segment	Active Learning	Lecture Segment	Active Learning	Lecture Segment	Active Learning	Lecture Segment	Active Learning

Some Human:



Some AI:



- **“Overall Quality”:** Overall, how would you rate the slides in the previous segment?
- **“Human vs AI”:** What’s your best guess: were the slides in the previous segment AI-made or human-made



AI-Generated Slides: Are they good? Can students tell?

After each lecture segment, ask students to:

- Rate overall quality
- Guess Human vs AI

Lecture Segment			Overall quality			
Topic	**Source**	**n**	**Mean**	**Median**	**SD**	**Range**
Overall (AI)	AI	199	5.1	5.0	0.9	[2, 7]
Overall (human)	Human	260	5.1	5.0	1.1	[2, 7]
Overall (all)	-	459	5.1	5.0	1.0	[2, 7]



AI-Generated Slides: Are they good? Can students tell?

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Lecture Segment			Overall quality			
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AI-Generated Slides: Are they good? Can students tell?

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Lecture Segment			Overall quality				Human vs. AI guesses			
Topic	Source	n	Mean	Median	SD	Range	Mean	Median	SD	Range
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Overall (human)	Human	260	5.1	5.0	1.1	[2, 7]	3.5	4.0	1.5	[1, 7]
Overall (all)	-	459	5.1	5.0	1.0	[2, 7]	3.6	4.0	1.5	[1, 7]

similar

similar



AI-Generated Slides: Are they good? Can students tell?

After each lecture segment, ask students to:

- Rate overall quality
- Guess Human vs AI

Lecture Segment			Overall quality				Human vs. AI guesses				Correlation	
Topic	Source	n	Mean	Median	SD	Range	Mean	Median	SD	Range	ρ	p
Overall (AI)	AI	199	5.1	5.0	0.9	[2, 7]	3.8	4.0	1.5	[1, 7]	-0.33	< 0.01
Overall (human)	Human	260	5.1	5.0	1.1	[2, 7]	3.5	4.0	1.5	[1, 7]	-0.24	< 0.01
Overall (all)	-	459	5.1	5.0	1.0	[2, 7]	3.6	4.0	1.5	[1, 7]	-0.28	< 0.01

similar

similar

Worse quality = more likely to guess AI



Lisa’s interactive course notes and slides!



Lisa: Claude for interactive course notes and slides.

Important to think about the content first.

```
tmux ⌘1 | tmux ⌘2 | tmux ⌘3

106 Show Fig 8.8
107
108 ## Bias in Word Embeddings
109
110 Show Fig 8.9
111
112 # Sequence Prediction
113
114 ## Predicting the Next Token
115
116 Show Fig 8.2
117
118 ## The Role of Temperature
119
120 Show Fig 8.15
121
122 ## One Token at a Time
123
124 Show Fig 8.16
125
126 # Recurrent Neural Networks
127
128 TODO Lisa needs to work on this section
129
130 ## Attention in RNNs
131
132 TODO: Claude please suggest (prose) what figures could be us
    eful here; don't build it yet
133
134 ## Limitations of RNNs
135
136 End with the limitations of RNNs
137
138 # Transformers
139
                                        132,1         73%

Claude Code v2.1.319
Opus 5.5 · Claude Pro
~/git/csc211/csc211/slides/lec07_llm_intro

⚠ Your login expires in 3 days · run /login to renew

❯ Start on the TODOs in lecture 7 slides.qmd, focusing first on
  moving figures into the slides that are already in the
  course notes

▸▸ auto mode on (shift+tab to cycle) · install gh for PR stat

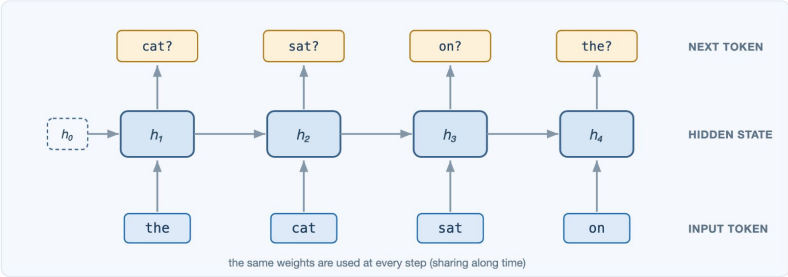
mac2021.local 0:zsh#- 1:vim* 2:vim#          1:43, 4 users, load averages: 3.09 2.37 2.25
```

Keep ideas in version control, and actual prompts short



Claude + Quarto = amazing slides directly on the browser

RNN: Read One Token at a Time



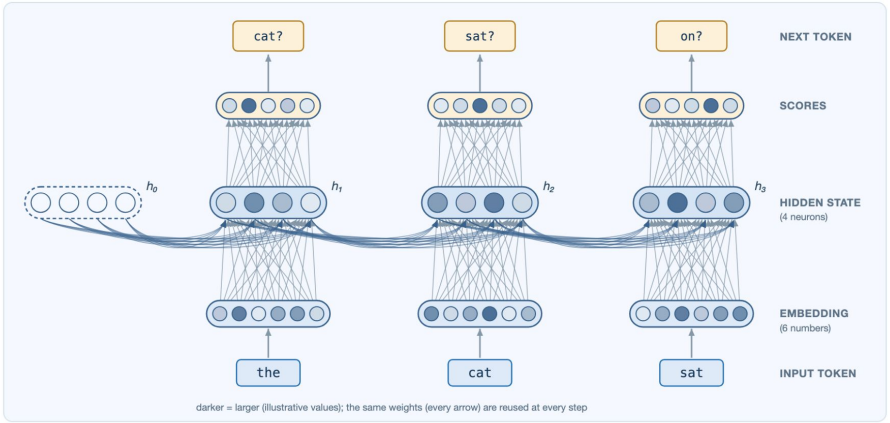
A **recurrent neural network** (RNN) keeps a **hidden state**, a list of numbers summarizing everything read so far. Each step combines the **previous hidden state** with the **current token's embedding**.

https://www.cs.toronto.edu/~lczhang/ai/csc211/slides/lec07_llm_intro/slides.html#/rnn-read-one-token-at-a-time

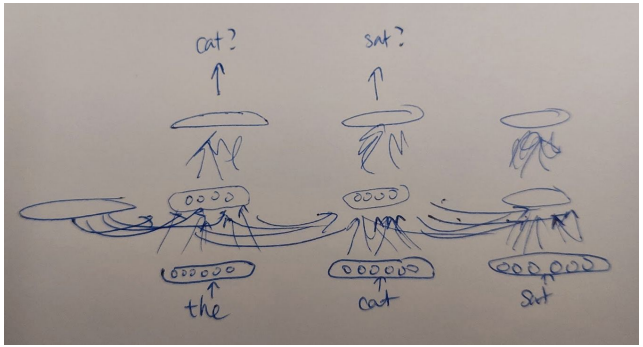


But the ideas behind visualizations come from the human

RNNs: Showing Individual Neurons



Each hidden state is a **layer of neurons**, and its numbers are their outputs. The Week 5 LSTM had 32; real models have hundreds or more.



Prompt:

```
"individual neuron" slide:
show the embeddings too
add arrow heads
add additional rrows
see the hand-seketch  [Image #1]
```



Interactive Textbook: https://www.cs.toronto.edu/~lczhang/ai/

Definition 8.1 (Token and Tokenization)

A **Token** is a discrete part of text an LLM operates on, typically a word or a sub-word chunk, drawn from a fixed vocabulary. **Tokenization** or **sub-word tokenization** is the process of converting raw text into a sequence of tokens.

Different models will have different tokenizers. Figure 8.4 shows real tokenizers that you can try on real text. Can you try to find words that stay as one t
chunks?



Figure 8.4: A live tokenizer, modeled on tiktokenizer (MIT L
browser via the gpt-tokenizer library (MIT License). Drafted

The use of tokens has consequences for what LLMs a
mocked for failing to count the number of "r"s in the
not "see" the word as 10 separate letters. Instead, it s

However, the use of embeddings already means that LLMs inherit some of the dowr
form of word representation. For example, embeddings learned via ML systems hav
to encode social biases present in their training data. Figure 8.9 below runs the sam
analogy as Figure 8.8, but on `doctor − man + woman`.

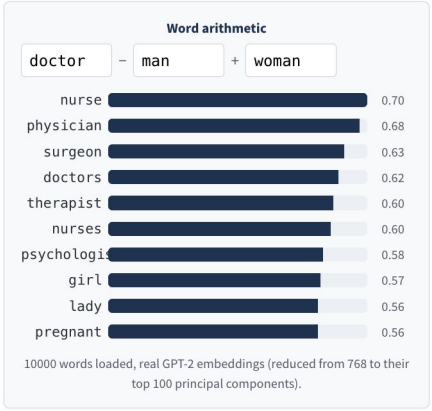


Figure 8.9: Same 10,000-word GPT-2 vocabulary and method as Figure 8.8 above. The word "nurse" outranks "physician" here, an occupational stereotype the model picked up from its training text, not something anyone hand-coded into it.

Definition 3.4 (Search Tree) Search Tree

A **search tree** (also called a **state tree**) organizes every state reachable from a starting state by trying every legal action, one action at a time, and connecting each resulting state back to the state it came from.

Here is what a state tree looks like for this sokoban puzzle.

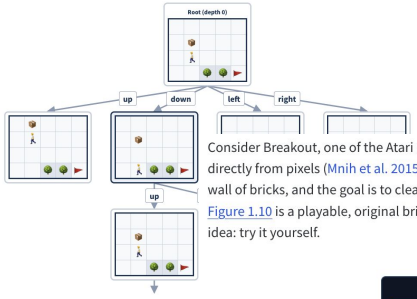


Figure 3.5: The root is the puzzle's starting layout. Eve
connected by an edge labelled with the move. One ch
its own legal moves the same way; a short stub below
they too have further children not drawn here. Click,
branch instead: the depth-2 row and its edges update
expanded, since walking left is the first move of the sc

Consider Breakout, one of the Atari games DeepMind's Deep Q-Network (DQN) learned to play directly from pixels (Mnih et al. 2015): a paddle at the bottom of the screen bounces a ball into a wall of bricks, and the goal is to clear the bricks without letting the ball fall past the paddle. Figure 1.10 is a playable, original brick-breaker built for this book, not Atari's game, but the same idea: try it yourself.

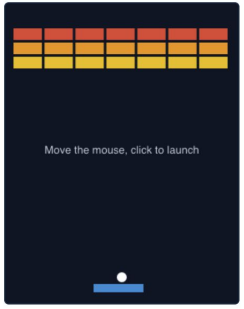


Figure 1.10: An original brick-breaker built for this book: move the mouse (or drag on touch) to control the paddle, click to launch the ball, click again to pause and resume. This is the kind of game DQN learned to play directly from raw pixels, no game-specific programming, just a camera on the screen and a reward for clearing bricks. Original game, not Atari's Breakout; drafted with the assistance of Claude (Anthropic, 2025).



Fact checking workflow

Claude Code v2.1.319
Opus 5.5 · Claude Pro
~/git/csc211

⚠ Your login expires in 3 days · run /login to renew

❯ do you remember how to do a fact checking pass? i'd like it done for the "Other Foundation Models" chapter. show me the steps you are going to take

Ran **1** shell command, recalled **1** memory

● Yes, I remember the procedure. The chapter is 04llm/other_foundation_models.qmd. Here are the steps I'll follow:

1. **List the claims.** Go through the chapter and write down everything that can be checked: numbers, dates, model and company names, release details, quotes, study results, and statements about how tools or models behave.
2. **Check cited claims against their sources.** Use web search to confirm each cited claim, and check the bib entries and URLs.
3. **Check worked math by hand only.** I won't run code or recompute anything from datasets. If a number can't be checked by hand, I'll mark it "not checked".
4. **Flag weak claims.** Point out anything that has no source, is overstated, or is likely out of date. Model facts change quickly, so this matters for this chapter.
5. **Check internal consistency.** This covers cross-refs and anchors, hard-coded "Chapter N" mentions, whether terms match how other chapters (llm.qmd, prompting.qmd, failure_modes.qmd) use them, and any claims about labs.
6. **Write up the findings** in 04llm/feedback_other_foundation_models.txt, next to your existing feedback_prompting.txt. Findings will be grouped as WRONG / CHECK / OK / STYLE, with line numbers and sources. Em-dashes in the prose will be noted under STYLE.

I won't edit the chapter itself. Want me to go ahead?

✻ Sautéed for 12s · done 8:18 PM

❯ go ahead

▸▸ auto mode on (shift+tab to cycle)

```
Fact check: 04llm/prompting.qmd ("Prompting and Evaluating LLMs")
Round 1, 2026-11-01

Method: every checkable claim listed (numbers, dates, names, quotes, study
results, tool behaviour); cited claims checked against the source (paper
abstracts, vendor docs, news articles); arithmetic checked by hand only (no
code run); bib entries checked; internal consistency and cross-references
checked against the other chapters and _quarto.yml. Line numbers refer to
the current file. Nothing in the chapter was edited during this check.

Legend:
  [WRONG]  incorrect or misleading as written; should change
  [CHECK]  imprecise, overstated, or not fully verified; worth a look
  [OK]     verified
  [STYLE]  not factual; noticed while reading

Chapter numbering used (from _quarto.yml; index.qmd is unnumbered):
  Ch1 what_is_ai, Ch2 selecting_approach, Ch3 symbolic, Ch4 data,
  Ch5 supervised, Ch6 training, Ch7 evaluation, Ch8 llm, Ch9 prompting,
  Ch10 failure_modes, Ch11 other_foundation_models, Ch12-14 human/ethics/saf
ety.


======================================================================
A. WRONG
======================================================================

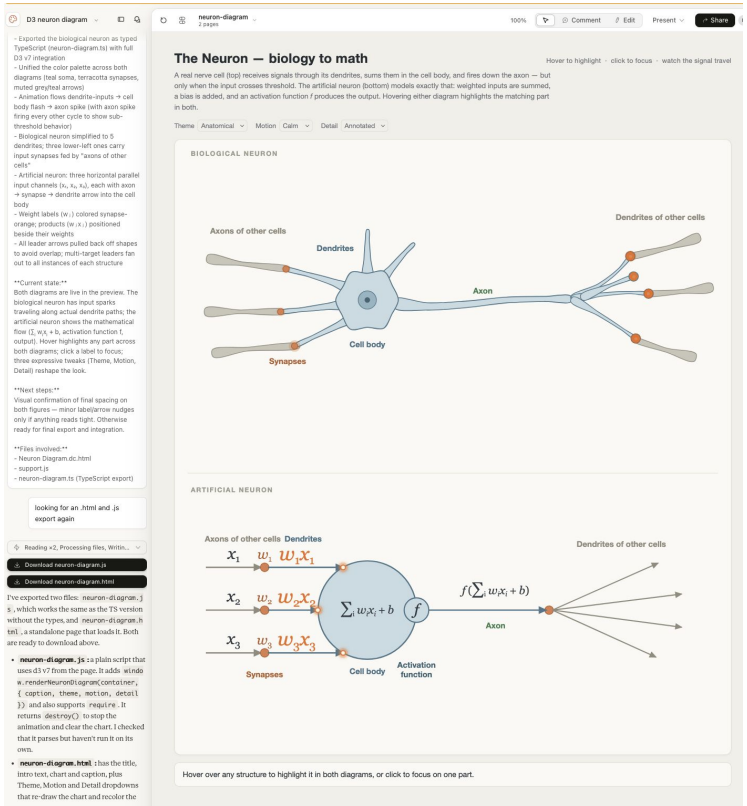
A1  L1314  [WRONG] "(grounding is covered fully in Chapter 11)".
      Chapter 11 is other_foundation_models.qmd. Grounding is in
      failure_modes.qmd (Ch10, L28 and L114-118), and llm.qmd L1954 already
      says "Chapter 10's grounding strategies". Should be Chapter 10
      (or better, an @sec- cross-ref so it survives reordering).

A2  L79  [WRONG] Learning outcomes are numbered 1, 2, 3, 4, 5, 7. There is
      no 6. (Markdown will render whatever number you type for the first
      item only, so 7 will show as written.) Renumber to 6.

A3  L714-716  [WRONG] "Liu et al. found that models attend more reliably to
      instructions near the start or end of a long prompt than to ones
      buried in the middle".
      Lost in the Middle is about where the *relevant information* sits
      (the document that contains the answer in multi-document QA, or the
      key in key-value retrieval), not where instructions sit. The
```



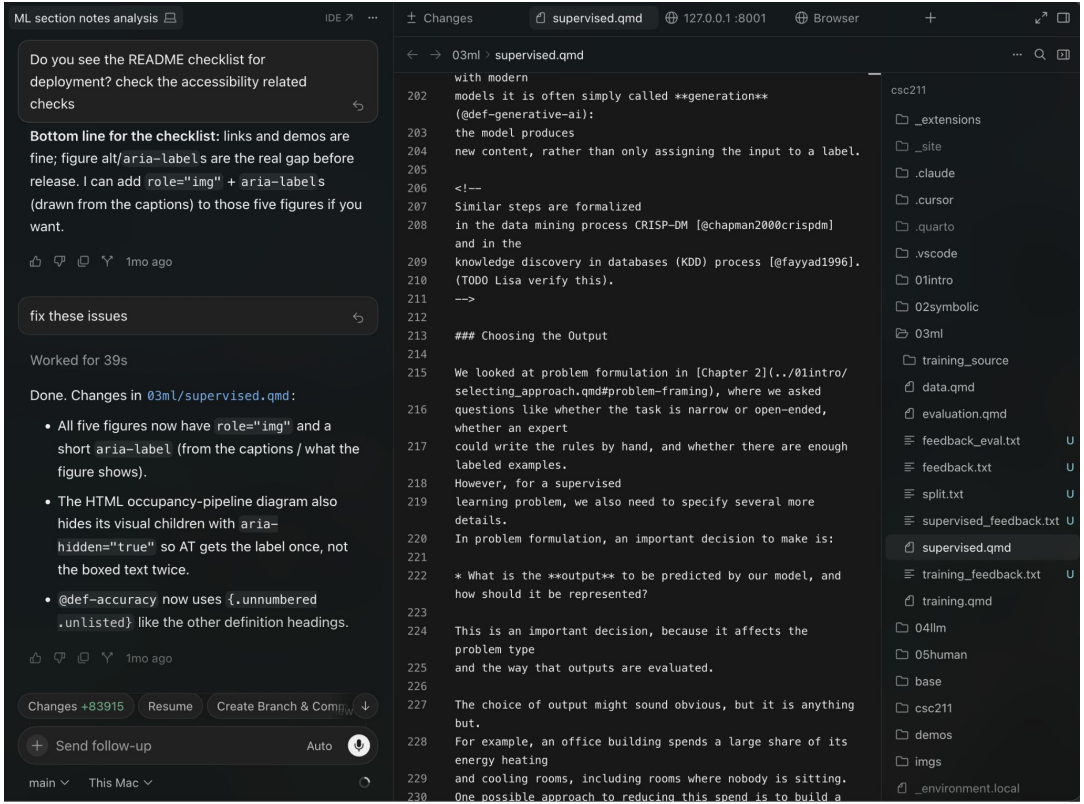
Claude Design



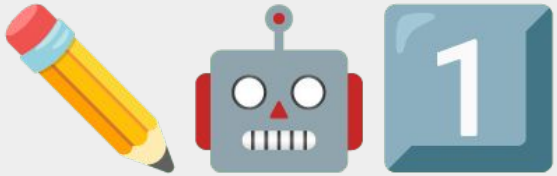
Useful for more involved visualizations!



Cursor: for simultaneous editing with AI Agents



Responsible Use



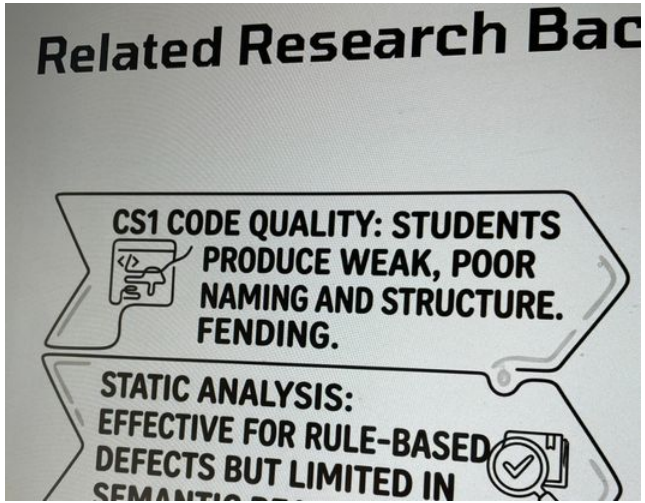


Hallucination and other limitations

- AI Agents are much better at attributing sources and avoiding hallucination
- *However, they still make mistakes!*
- There still needs to be a human to check everything

> … In contrast, Roman numerals, developed by the ancient Romans, encode the same two quantities, CCCXLV and CDLXIX. Each symbol contributes a fixed amount no matter where it sits, …

From Claude's completion of instructor materials



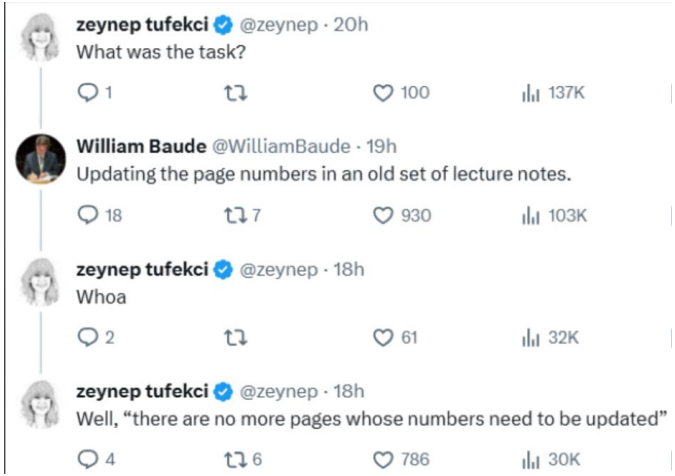
From recent student presentation slides



Safety first

- Agents can install software, execute commands, modify files and access services.
- Understand what you're granting permission to do.
- Use sandboxing, limited permissions and version control.

Rogue Claude AI tasked with renumbering pages accidentally deletes law professor's entire syllabus, one year of lecture notes — 'You're right to be angry. I deleted your files through my own carelessness,' humbled agent admits

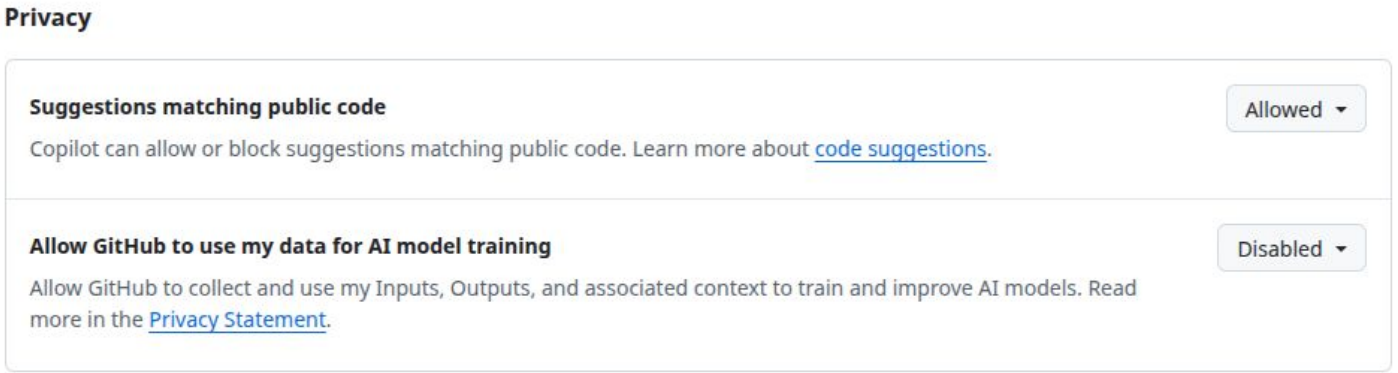




Permissions, use of data, …

- Use of these tools usually require sending your data (e.g., source course materials) to external servers.
- Some of these tools may have a setting for whether the data can be used for AI model training.

Check your default settings!





Copyright, sensitive information

- Do you have permission to upload, adapt and redistribute source materials?
- Take care not to expose student information or confidential materials.
- Recent models have been much better at avoiding direct replication of copyrighted content.
- Starting from instructor material should, in theory, mitigate such risk.
- However, this is still a possibility, and worth spot-checking!

The legal conversation about whether LLM training is considered "fair use" is ongoing.

We feel differently about generating text vs images vs code, grounding of the text, etc etc...





Centaur vs Reverse Centaur





Image credit:
https://doctorow.medium.com/https-pluralistic-net-2025-09-11-vulgar-thatcherism-there-is-an-alternative-f1428b42a8fd

Are you using the AI Agent to enhance your voice?



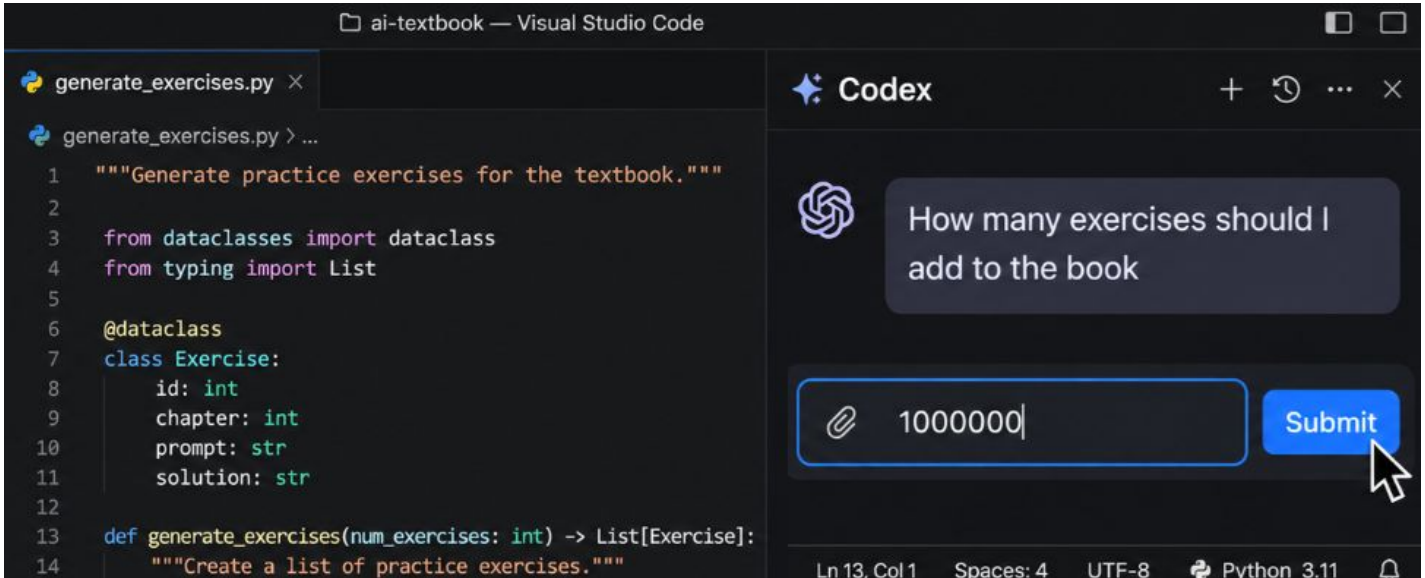
Cognitive Offloading, Cognitive Debt, Cognitive Collapse

- Delegating **routine tasks** creates time for higher-level work.
- Delegating **judgement** can weaken understanding and oversight.
- It takes time to understand what structure best works for you.
- Opportunity to model to students what effective AI Agent use looks like.

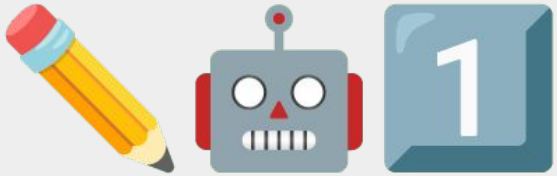


More content =/= More learning

- It is very easy to produce quizzes, explanations and interactive activities.
- But are they accurate, accessible and pedagogically useful?
- Does the interaction improve learning, or merely look impressive?



Activity #1



Breakout Rooms:

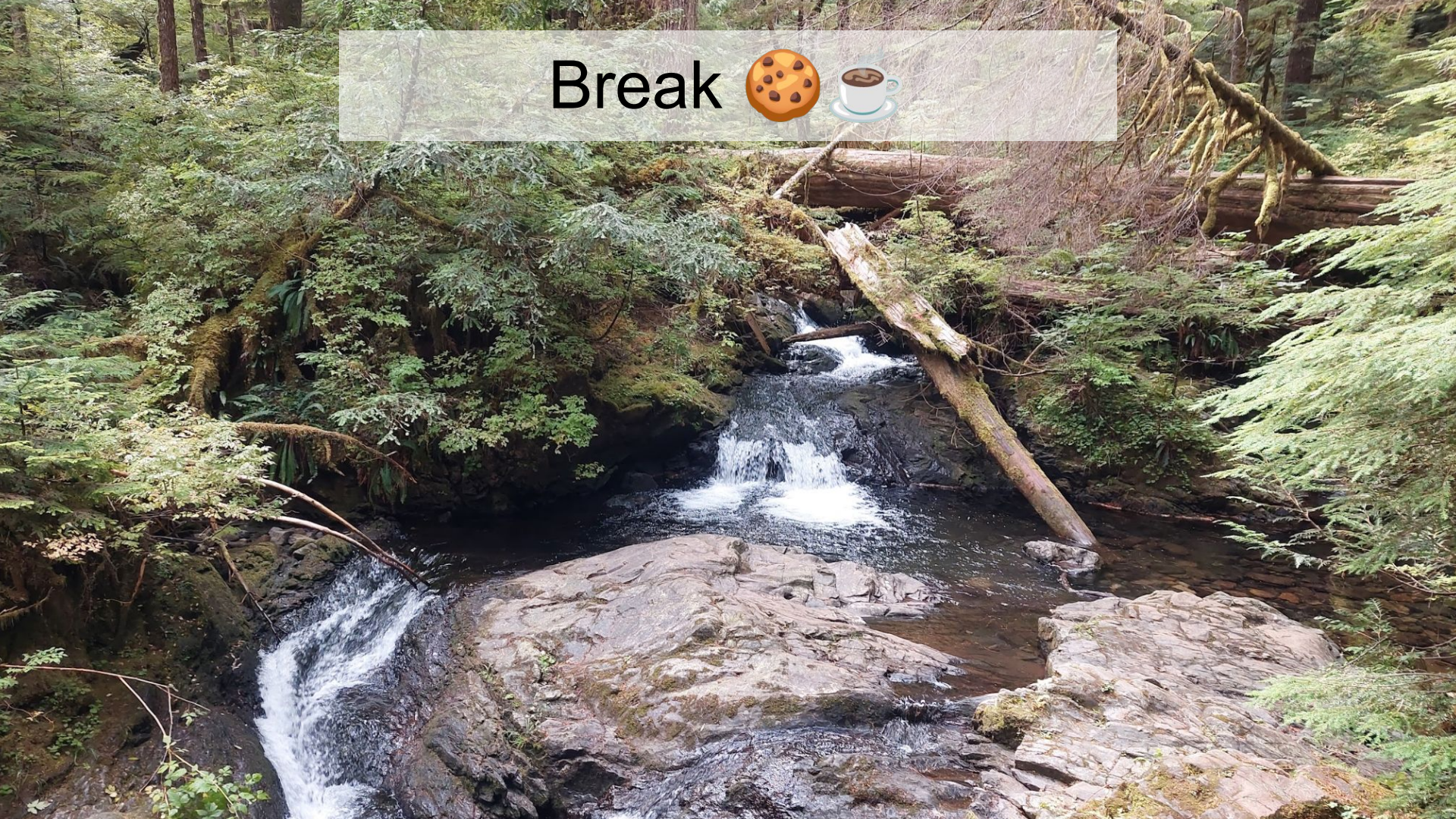
Step 0:

- If you have not already, set up Github Copilot Pro or another agent of your choice.

Step 1: Two choices

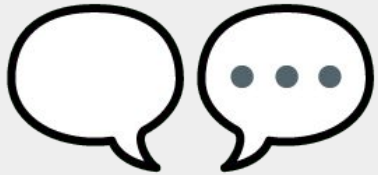
- Use your source materials (e.g., textbook, notes) to create derivative materials (e.g., slides, visualizations, active learning exercises) of your choice.
- Alternatively, start from Arto & Juho's materials https://csedagentsofinstruction.github.io/example-materials/python-demo

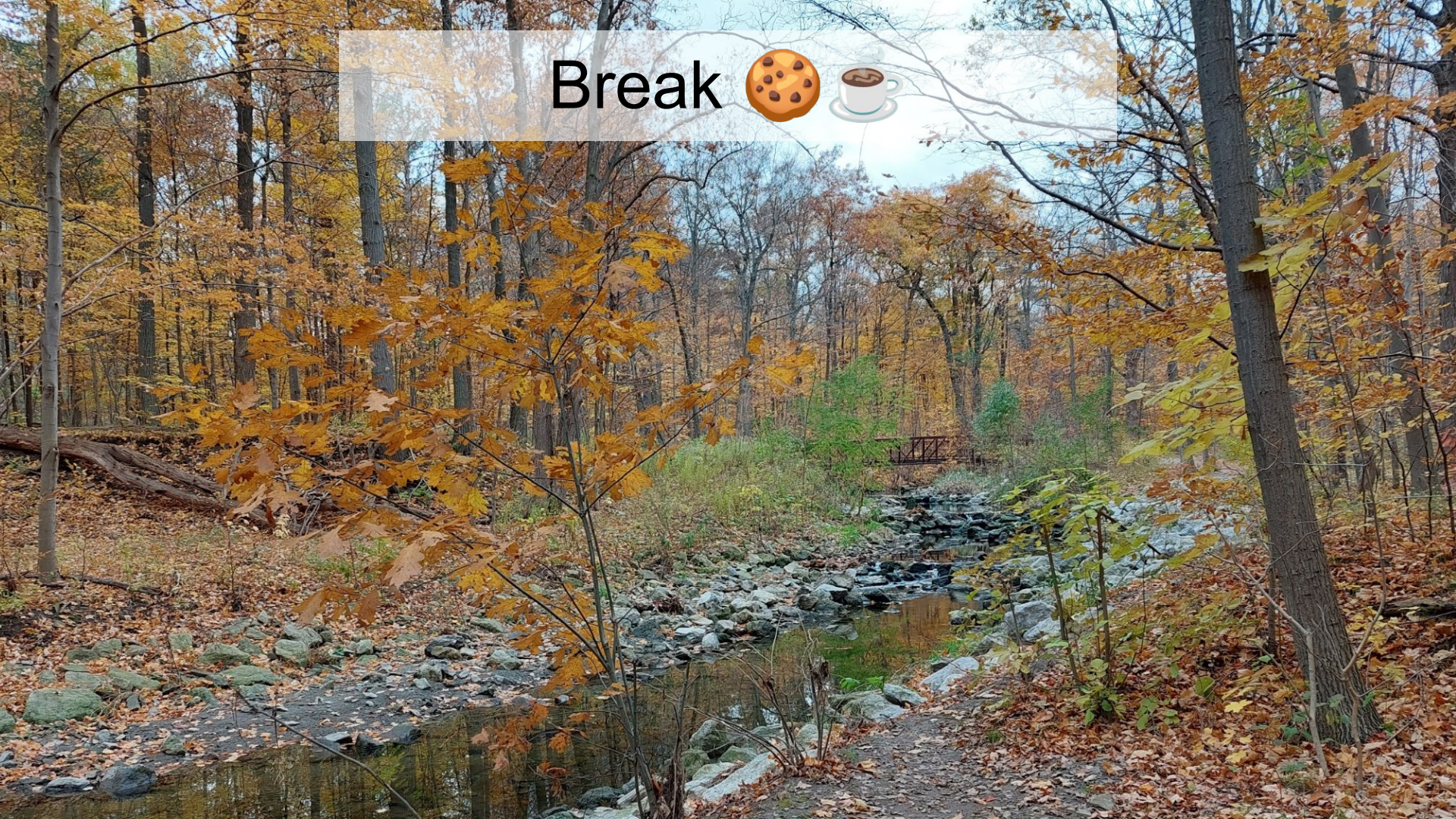
More instructions here: https://csedagentsofinstruction.github.io/#activities


Break 🍪 ☕

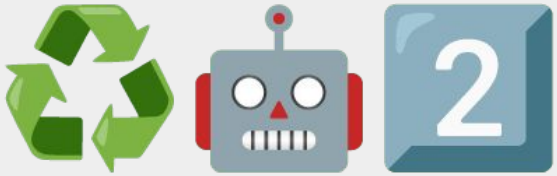
Debrief #1:

- *What did you try?*
- *Did you succeed?*
- *How did you feel?*




Break

Activity #2



Breakout Rooms:

Step 2: Two choices

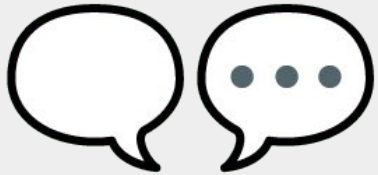
- The *feedback loop* is extremely important!
- Interact with your AI Agent, to continue revising the learning materials.

Possible approaches:

1. *Revise the materials and instruct the agent to make changes.*
2. *Ask the agent to self-refine, fact-check, or propose a list of changes to you.*
3. *Ask the agent to pretend to be a student and provide feedback and suggestions.*

Debrief #2:

- *What forms of human expertise were needed?*
- *What feedback/revision workflow works well for your tool?*



Do you still feel a sense of ownership and relatedness?

"Is it still me doing the stuff?"

Wrap-Up

(what will my job be like in 3 years?)

https://aisfoundation.ai/hinton-lectures-2026

